



Research paper

Modeling of hysteresis curve and study of torsional rigidity and lost motion characteristics of the RV reducer

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ABSTRACT

Torsional rigidity and lost motion are critical performance indicators for evaluating the load capacity, motion stability, and positioning accuracy of the rotating vector (RV) reducer. This paper presents a dynamic modeling method for the RV reducer's hysteresis curve and evaluates its torsional rigidity and lost motion based on the hysteresis curve. Firstly, the method provides the contact dynamic model of the RV reducer, including the influence of geometric errors in components. Secondly, the simulation control strategy for torque loading is developed based on the hysteresis curve test scheme. Besides, the validity of the calculation results is verified by experiment. Moreover, the effects of contact stiffness, bearing preload and clearance, and geometric errors of components on the hysteresis curve, as well as the torsional rigidity and lost motion, are discussed in detail. The results show that support bearing contact stiffness has the most significant influence on the torsional rigidity; radial clearance of the swivel arm bearing has a significant impact on the lost motion; and geometric errors of components will reduce the torsional rigidity and increase the lost motion.

1. Introduction

The RV reducer is widely used in the precision transmission of industrial robot joints, offering advantages such as high transmission precision, high load capacity, and high transmission efficiency. Recently, many scholars have conducted extensive research on the transmission error [1–4], contact characteristics [5–10], vibration characteristics [11–14], transmission efficiency [15–18], and cycloid gear tooth modification [19–22] of the RV reducer. However, there has been little research on the hysteresis curve characteristics of the RV reducer. The torsional rigidity and lost motion obtained from the hysteresis curve are significant performance indicators of the RV reducer, which directly influence the working performance of industrial robots, including dynamic performance, load capacity, and positioning accuracy.

Insufficient torsional rigidity in the RV reducer will result in abnormal vibrations during operation and excessive elastic deformation of internal components, diminishing the load capacity and motion accuracy of the RV reducer and thereby directly impacting the operational performance and working quality of industrial robots. In recent years, some scholars have conducted research on the stiffness characteristics of RV reducers. E et al. [23] proposed a method for calculating the meshing stiffness of the cycloid-pin pair of the RV reducer under conditions with localized faults in the cycloid gear and pin tooth. Ren et al. [24] proposed a new modification method for cycloid gear, and the meshing stiffness of the cycloid-pin pair was calculated by the potential energy method. Li et al. [25]

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proposed a calculation method for the single-tooth contact stiffness and the multi-tooth torsional stiffness of the cycloid-pin pair, taking into account tooth modification and eccentricity error. Based on static test results, Kumar et al. [26] proposed a semi-analytical model for calculating the torsional flexibility (reciprocal of torsional rigidity) of a single-stage cycloidal reducer. Liu et al. [27] designed a new type of cycloid transmission mechanism with a double-enveloping cycloidal tooth profile and established the calculation model for its meshing stiffness and torsional rigidity. Song et al. [28] proposed a torsional rigidity analysis model for the cycloid-pin pair that considers the load torque and cycloid tooth modification. This model utilizes finite element simulation methods to determine the maximum deformation of the meshing position of the cycloid-pin pair. Liu et al. [29] proposed a calculation model for the torsional rigidity of the cycloid-pin pair by combining fractal contact theory and the finite element method. Zhang et al. [30] developed a dynamic model of RV reducers considering wear. They obtained the degradation law of the equivalent torsional rigidity of the cycloid-pin pair during the wear update process. Xie et al. [31] presented a dynamic model of the non-standard angular contact ball bearing-planetary frame rotor system in the RV reducer. They studied the moment rigidity of the RV reducer. Cheng et al. [32] proposed a method for calculating the dynamic reducer torsional rigidity and a method for testing dynamic torsional rigidity based on transmission error. Chang et al. [33] proposed a dynamic model of the involute variable-thickness RV reducer based on the Lagrange method and proposed a method for calculating torsional rigidity by using the input angle, output angle, and output torque. Zhou et al. [34] established a torsional rigidity analysis model for the RV reducer based on the stiffness-topology relationship and compared the effects of various structural parameters and stiffness parameters on torsional rigidity. However, this study did not analyze the impact of supporting bearing stiffness on torsional rigidity. Li et al. [35] developed an analytical model for the time-varying torsional rigidity of the RV reducer, which requires the use of finite element simulation methods to obtain the time-varying radial stiffness of the crankshaft and support bearings. Wang et al. [36] studied the torsional rigidity fluctuation characteristics of the RV reducer based on experimental data and constructed the torsional vibration equation of the RV reducer.

Lost motion of the reducer refers to the phenomenon that the output end lags behind the input end in motion when the input end of the reducer rotates in reverse [37]. Due to the existence of lost motion, when the input end of the reducer rotates in reverse, the output end cannot respond synchronously. However, the RV reducer in industrial robot joints typically requires reciprocating motion. The presence of lost motion can significantly impact the repetitive positioning accuracy and feedback control of industrial robot systems. Currently, some scholars have also researched the lost motion of the reducer. Sun et al. [38] designed a new type of single-stage cycloid reducer. They studied the initial tooth side clearance of the cycloid-pin pair caused by cycloid tooth modification and proposed a composite modification method of “negative isometric modification + negative offset modification” to reduce lost motion. Tran and Pham et al. [39,40] established a finite element simulation model of a single-stage cycloid reducer. Based on the simulation method, they studied the hysteresis characteristics of the reducer and investigated the effects of tolerances on the lost motion, backlash, and torsional rigidity of the reducer. Wang et al. [41] developed an analysis model for the lost motion of the RV reducer, taking into account manufacturing and installation errors through geometric analysis methods. They then analyzed the influence of the clearance between the transmission components of the RV reducer on the lost motion. Zhang et al. [42,43] proposed an equivalent mechanism model with parallel linkages for analyzing lost motion in multi-stage planetary gear transmission systems. They analyzed the generation mechanism of lost motion and studied the effect of manufacturing errors on lost motion in planetary gear transmission systems. Cheng et al. [44] proposed two innovative loading techniques for the lost motion test: equal position gradient loading and uniform speed loading. These techniques address the limitation of the traditional equal torque gradient loading technique, which is constrained by the servo motor’s loading resolution in the lost motion test of small reducers.

Currently, only a few studies have investigated the overall torsional rigidity and lost motion performance of RV reducers. Furthermore, most previous studies on the torsional rigidity of RV reducers have ignored the influence of geometric errors in transmission components. Most importantly, the test results of the hysteresis curves serve as the primary basis for evaluating the torsional rigidity and lost motion of the RV reducer in engineering. However, currently, there is a relative lack of theoretical research on the characteristics of the hysteresis curve of the RV reducer, and few studies can analyze both the overall torsional rigidity and lost motion of the RV reducer simultaneously. Aiming at the limitations of existing research, this paper proposes a dynamic modeling method for the hysteresis curve of the RV reducer and evaluates the torsional rigidity and lost motion of the reducer based on the hysteresis curve. The remainder of the paper is organized as follows: Section 2 introduces the method for evaluating the torsional rigidity and lost motion of the RV reducer based on the hysteresis curve, as well as the process for generating the hysteresis curve. Section 3 introduces a contact dynamic modeling method for the RV reducer, including methods for calculating the contact stiffness of planetary gear pair, cycloid-pin pair, and bearing roller raceways, as well as the contact modeling process. Section 4 presents a simulation control strategy for the torque loading of the RV reducer and simulates the hysteresis curve of the RV reducer through the dynamic model. This section then evaluates the torsional rigidity and lost motion of the RV reducer at different working positions, based on the hysteresis curve, and conducts experimental verification. Section 5 discusses the effects of transmission interface contact stiffness, bearing preload and radial clearance, component geometric errors, and loading rate on the hysteresis curve, torsional rigidity, and lost motion of the RV reducer. In the last section, the main conclusions from this study are presented.

2. Hysteresis curve of RV reducer

The reducer hysteresis curve is a characteristic curve that describes the hysteresis phenomenon of the reducer. It reflects the nonlinear relationship between the load torque and torsional deformation at the output end of the reducer under cyclic load. The hysteresis characteristics of the reducer are manifested as inconsistencies between the system input and its corresponding output in the forward and reverse strokes. The hysteresis characteristics of reducers are affected by factors such as geometric errors, stiffness, and friction [45]. Due to transmission clearance caused by geometric errors, elastic deformation of transmission components caused by

loads, and resistance to motion caused by friction, when the direction of the input end of the reducer changes, the motion response of the output end lags behind that of the input end.

In engineering, the hysteresis curve test results are commonly used to evaluate the torsional rigidity and lost motion of the RV reducer. The method of evaluating the torsional rigidity and lost motion of the RV reducer based on the hysteresis curve is shown in Fig. 1. The inclination of the hysteresis curve reflects the torsional rigidity of the reducer. The ratio of a to b is the torsional rigidity of the reducer. In Fig. 1, a denotes the torque segment near the rated torque T_r on the hysteresis curve, and b represents the corresponding torsional angle segment on the hysteresis curve. At $\pm 3\%$ of the rated torque, the difference in the torsional angle at the midpoint of the hysteresis curve width is defined as the lost motion.

In the hysteresis curve test of the RV reducer, the input shaft of the reducer needs to be fixed, and the output end needs to be loaded and unloaded in cycles. Furthermore, the loading and unloading of torque must be carried out in a step-by-step manner. As shown in Fig. 2, in order to obtain a closed reducer hysteresis curve, the application process of the output end load torque needs to be divided into five stages: 1) Perform stepwise loading from no load to rated torque T_r in the forward direction; 2) Perform stepwise unloading to no load; 3) Perform stepwise loading in the reverse direction to rated torque T_r ; 4) Perform stepwise unloading to no load; 5) Perform stepwise loading in the forward direction to rated torque T_r . The load torque can cause torsional deformation in the reducer. During the above process, record the torque and torsional angle at the output end of the reducer synchronously. With torque as the horizontal coordinate and torsional angle as the vertical coordinate, the hysteresis curve of the reducer can be plotted. In the loading process shown in Fig. 2, the closed curve obtained in stages 2 to 5 represents the hysteresis curve.

3. Contact dynamic model of the RV reducer

Based on the theories of contact mechanics and multi-body dynamics, a contact dynamics model for RV reducer was established. According to the test method of hysteresis curve, the hysteresis curve of the reducer can be generated by fixing the input gear and applying cyclic load torque at the output end in the dynamic model. Fig. 3 illustrates the contact dynamic model of the RV reducer. In the RV reducer, the planetary gear is connected to the crankshaft through the involute spline. The multi-teeth engagement characteristic makes spline with high rigidity. Moreover, it has been demonstrated in the literature [4] and [34] that the planetary gear pair in the high-speed stage exerts minimal influence on the performance of the RV reducer, such as torsional rigidity and transmission accuracy. Therefore, although there is a tiny tooth side clearance between the internal and external splines, since the planetary gear pair is located in the high-speed stage, the tooth side clearance has a very small impact on the performance of the reducer and can be ignored. In addition, the output disc and the compression disc are connected with high rigidity through screws and tapered pins. Therefore, in the model, it is assumed that the planetary gear and the crankshaft, as well as the output disc and the compression disc, are the same components, respectively. The pin teeth are considered fixed. The model consists of six transmission components, including an input gear shaft, two planetary gear and crankshaft assemblies, two cycloid gears, and an output assembly consisting of an output disc and a compression disc. The impacts of the swivel arm bearing and the support bearing are also considered in the model. The coordinate system XOY is the generalized coordinate system of the transmission system, and the output disc rotates around the Z axis. To simulate the hysteresis curve of the RV reducer, it is necessary to fix the input gear shaft and apply variable torque to the output disc in the dynamic model. The load torque is represented by T_{load} .

The above transmission system has a total of 18 generalized coordinates, represented as:

$$\mathbf{q} = (x_k, y_k, \theta_k, x_{d1}, y_{d1}, \theta_{d1}, x_{d2}, y_{d2}, \theta_{d2}, x^A, y^A, \theta^A, x^B, y^B, \theta^B, x_q, y_q, \theta_q)^T \quad (1)$$

where (x_k, y_k, θ_k) is the generalized coordinate of the input gear shaft, $(x_{d1}, y_{d1}, \theta_{d1})$ and $(x_{d2}, y_{d2}, \theta_{d2})$ are the generalized coordinates of the two sets of planetary gear and crankshaft assemblies, (x^A, y^A, θ^A) and (x^B, y^B, θ^B) are the generalized coordinates of the two cycloid gears, (x_q, y_q, θ_q) is the generalized coordinate of the output assembly.

The mass matrix of the transmission system can be expressed as:

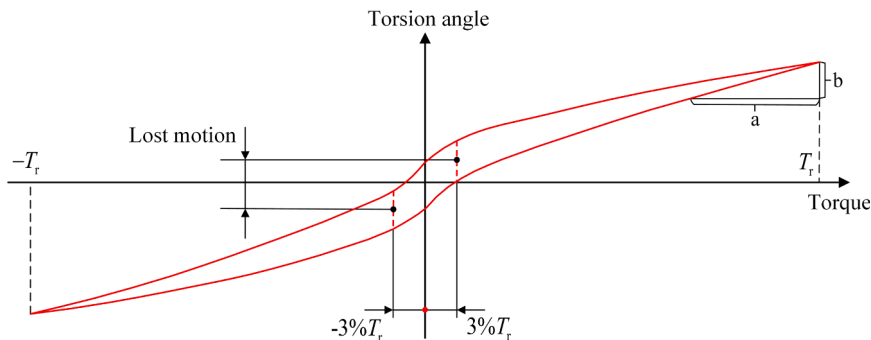


Fig. 1. Hysteresis curves of the RV reducer.

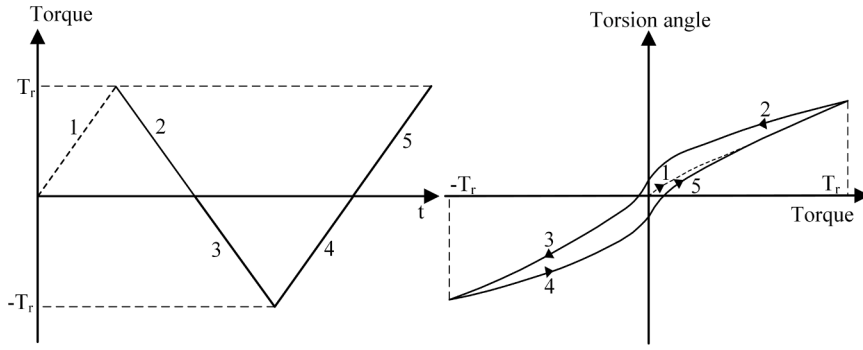


Fig. 2. Hysteresis curve generation process.

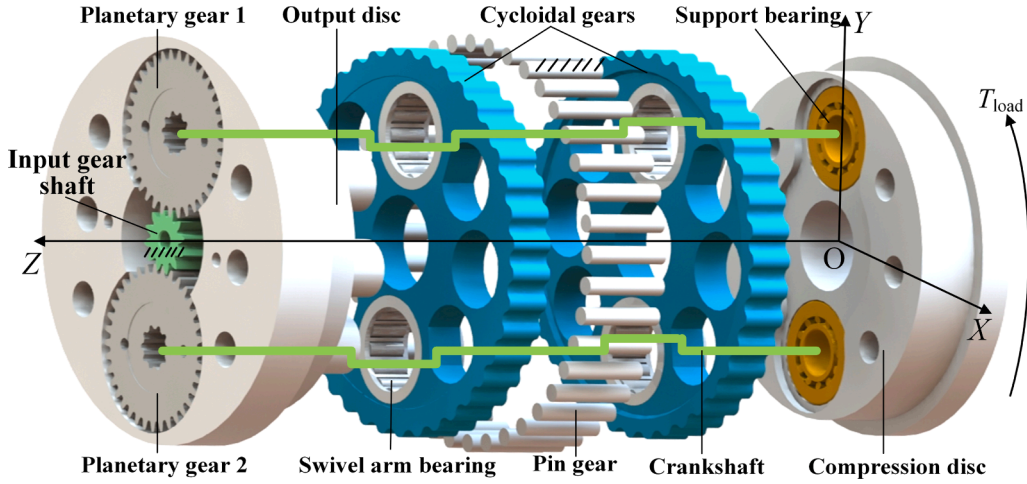


Fig. 3. Contact dynamic model of RV reducer.

$$\mathbf{M} = \text{diag}(m_k, m_k, I_k, m_d, m_d, I_d, m_d, m_d, I_d, m_c, m_c, I_c, m_c, m_c, I_c, m_q, m_q, I_q) \quad (2)$$

where m_k and I_k are the mass and moment of inertia of the input gear shaft, respectively, m_d and I_d are the mass and moment of inertia of the planetary gear and crankshaft assembly, m_c and I_c are the mass and moment of inertia of the cycloid gear, m_q and I_q are the mass and moment of inertia of the output assembly.

The dynamic equation of the transmission system can be expressed as [46]:

$$\begin{pmatrix} \mathbf{M} & \Phi_q^T \\ \Phi_q & \mathbf{0} \end{pmatrix} \begin{pmatrix} \ddot{\mathbf{q}} \\ \ddot{\lambda} \end{pmatrix} = \begin{pmatrix} -(\Phi_q \dot{\mathbf{q}})_q \dot{\mathbf{q}} - 2\Phi_{qt} \dot{\mathbf{q}} - \Phi_{tt} \\ \mathbf{Q}^G \end{pmatrix} \quad (3)$$

where Φ_q is the Jacobian matrix of the constraint equation, which contains the motion constraint relationship of the input gear shaft and output disc of the RV reducer, the Jacobian matrix is the partial derivative matrix of the constraint equation vector with respect to the generalized coordinates vector, $\dot{\mathbf{q}}$ is the generalized velocity vector of the system. $\ddot{\mathbf{q}}$ is the generalized acceleration vector of the system, λ is the Lagrange multiplier, $\mathbf{Q}^G$ is the generalized force vector of the system, including gravity, load torque, and contact forces between components, Φ_{qt} is the time derivative of the Jacobian matrix of the constraint equation, Φ_{tt} is the second derivative of the constraint equation to time.

As mentioned above, during simulation of the hysteresis curve, it is necessary to fix the input gear shaft and apply torque to the output disc. Therefore, in the dynamic model, the three degrees of freedom of the input gear shaft need to be constrained, while only the rotational degree of freedom of the output disc is retained. Thus, the constraint equation of the transmission system can be expressed as:

$$\Phi = \begin{Bmatrix} x_k \\ y_k \\ x_q \\ y_q \\ \theta_k \end{Bmatrix} = 0 \quad (4)$$

3.1. Calculation of contact stiffness of transmission interface

The dynamic model constructed in this paper includes contact modeling for involute planetary gear pair, cycloid-pin pair, swivel arm bearing, and support bearing. Contact stiffness is a critical parameter in contact modeling and significantly influences torsional rigidity. Therefore, it is necessary to first ensure the accuracy of contact stiffness calculations.

3.1.1. Meshing stiffness of the gear pair

In this paper, the potential energy method is used to calculate the meshing stiffness of the gear pair. The potential energy method divides the stiffness of a gear system into four parts: Hertzian contact k_h , bending stiffness k_b , axial compression stiffness k_a , and shear stiffness k_s . As shown in Fig. 4, this paper simplifies the gear into a cantilever beam starting from the root circle based on the potential energy method. The stiffness of the above four parts can be calculated by the following formula [47]:

$$\frac{1}{k_h} = \frac{4(1-\nu^2)}{\pi EB} \quad (5)$$

$$\frac{1}{k_a} = \int_{\frac{\pi}{2}}^{\alpha} \frac{\sin^2 \beta}{EA_{y1}} \frac{dy_1}{d\gamma} d\gamma + \int_{\tau_c}^{\beta} \frac{\sin^2 \beta}{EA_{y2}} \frac{dy_2}{d\tau} d\tau \quad (6)$$

$$\frac{1}{k_b} = \int_{\frac{\pi}{2}}^{\alpha} \frac{[\cos \beta (y_\beta - y_1) - x_\beta \sin \beta]^2}{EI_{y1}} \frac{dy_1}{d\gamma} d\gamma + \int_{\tau_c}^{\beta} \frac{[\cos \beta (y_\beta - y_2) - x_\beta \sin \beta]^2}{EI_{y2}} \frac{dy_2}{d\tau} d\tau \quad (7)$$

$$\frac{1}{k_s} = \int_{\frac{\pi}{2}}^{\alpha} \frac{1.2 \cos^2 \beta}{GA_{y1}} \frac{dy_1}{d\gamma} d\gamma + \int_{\tau_c}^{\beta} \frac{1.2 \cos^2 \beta}{GA_{y2}} \frac{dy_2}{d\tau} d\tau \quad (8)$$

where B is the tooth surface contact width of the gear, E and ν are the elastic modulus and Poisson's ratio of the gear material, respectively, β is the force action angle, α is the pressure angle, τ_c is the pressure angle at the starting point of the involute curve, G is the shear modulus. The meanings and calculation methods of other parameters in the formulas above can be found in the literature [47].

Additionally, deformation of the gear base also affects the equivalent stiffness at the gear meshing line. Therefore, the calculation of gear pair meshing stiffness must also consider the stiffness of the base deformation. The method for calculating the stiffness of the base deformation is as follows:

$$\frac{1}{k_f} = \frac{\cos^2 \beta}{EB} \left[L^* \left(\frac{u_f}{S_f} \right)^2 + M^* \left(\frac{u_f}{S_f} \right) + P^* (1 + Q^* \tan^2 \beta) \right] \quad (9)$$

where parameters u_f , S_f , L^* , M^* , P^* , Q^* can be found in the literature [48].

The meshing stiffness of a single tooth pair can be calculated by the following formula:

$$K_{kd} = \frac{1}{\frac{1}{k_h} + \frac{1}{k_{b-k}} + \frac{1}{k_{a-k}} + \frac{1}{k_{s-k}} + \frac{1}{k_{f-k}} + \frac{1}{k_{b-d}} + \frac{1}{k_{a-d}} + \frac{1}{k_{s-d}} + \frac{1}{k_{f-d}}} \quad (10)$$

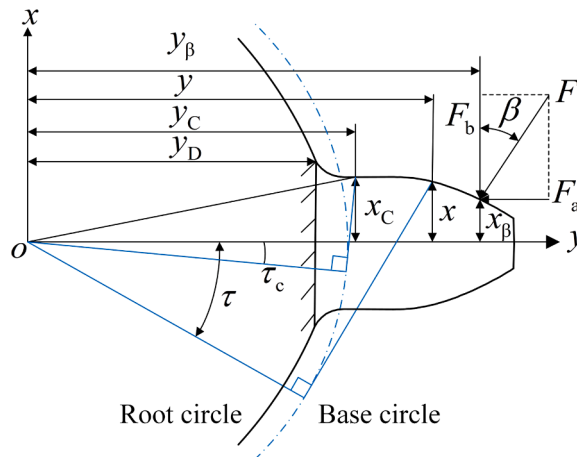


Fig. 4. Gear cantilever model.

where the subscript k represents the drive gear, the subscript d represents the driven gear.

When multiple tooth pairs mesh simultaneously, the meshing stiffness of the gear pair can be expressed as:

$$K_m = \sum_{i=1}^n K_{kd}^i \quad (11)$$

where n is the number of tooth pairs participating in meshing at a specific time.

3.1.2. Contact stiffness of the cycloid-pin pair

Fig. 5 illustrates the schematic diagram of the contact between the cycloid gear and the pin. Theoretically, the cycloid gear and pin tooth transmit motion through line contact. According to Hertz's contact theory, the contact between the cycloid gear and pin tooth can be equivalent to the contact between two cylinders. Assuming that the materials of the cycloid gear and pin tooth are the same, the contact stiffness between a cycloid gear tooth and a pin tooth can be expressed as follows [28]:

$$K_c = \frac{\pi B_c E_c |\rho_c|}{4(1 - \nu_c^2) \rho_z} \quad (12)$$

$$K_p = \frac{\pi B_c E_c r_{rp}}{4(1 - \nu_c^2) \rho_z} \quad (13)$$

$$K_{cp} = \frac{1}{\frac{1}{K_c} + \frac{1}{K_p}} = \frac{\pi B_c E_c r_{rp} |\rho_c|}{4(1 - \nu_c^2)(r_{rp} + |\rho_c|) \rho_z} \quad (14)$$

where K_c and K_p are the normal stiffnesses of a single cycloid tooth and pin tooth, respectively, B_c is the tooth surface contact width of the cycloid gear, ν_c is Poisson's ratio of the cycloid gear and pin tooth material, E_c is the elastic modulus of the cycloid gear and pin tooth material, ρ_c is the radius of curvature of each discrete point on the cycloid gear profile, ρ_z is the combined radius of curvature of the contact point between the pin tooth and the cycloid gear, it can be calculated by the following formula:

$$\rho_z = \frac{|\rho_c| r_{rp}}{|\rho_c| \pm r_{rp}} \quad (15)$$

In this equation, when $\rho_c < 0$, the convex part of the cycloid gear profile comes into contact with the pin tooth, “ $\pm$ ” takes the “+” sign; otherwise, it takes the “-” sign.

Through the above equations, it can be seen that the contact stiffness of the cycloid-pin pair depends on the curvature radius of the point of force on the cycloid gear profile. At different integration times during the dynamic calculation, the pin tooth comes into contact with different discrete points on the cycloid gear profile, so the contact stiffness between the pin tooth and the cycloid gear exhibits time-varying characteristics.

3.1.3. Contact stiffness between the bearing roller and raceway

The swivel arm bearing of the RV reducer adopts the needle roller bearing configuration with cylindrical rollers. The total contact stiffness between the cylindrical roller and the inner/outer raceways can be calculated as follows [49]:

$$K_g = [(K_i)^{-9/10} + (K_o)^{-9/10}]^{-10/9} \quad (16)$$

where the contact stiffness between the roller and the inner/outer raceways K_i and K_o can be calculated as follows:

$$K_\eta = 8.06 \times 10^7 I_g^{8/9}, (\eta = i, o) \quad (17)$$

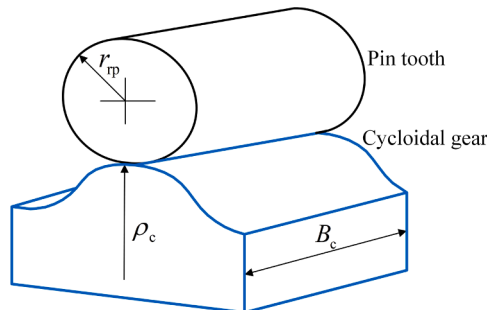


Fig. 5. Contact between the cycloid gear and pin tooth.

where l_g is the length of cylindrical rollers.

The supporting bearing of the RV reducer utilizes the tapered roller bearing configuration with tapered rollers. The total contact stiffness between the tapered roller and the inner/outer raceways can be calculated as follows [50]:

$$K_h = 8.06 \times 10^4 l_h^{8/9} \times [1 + c_i^{9/10} \cos(\alpha_e - \alpha_i)]^{-10/9} \quad (18)$$

where l_h is the length of tapered rollers, α_i and α_e are the contact angles of the inner and outer raceways, respectively. The formula for calculating c_i is as follows:

$$c_i = \frac{\sin(\alpha_e + \alpha_f)}{\sin(\alpha_i + \alpha_f)} \quad (19)$$

where α_f is the contact angle of the bearing flange.

There is a set of supporting bearings on both the output disc and the compression disc. However, the dynamic model presented in this paper is a planar model, which treats the output disc and the compression disc as a single component. Therefore, the support function of the two sets of bearings cannot be directly constructed. On the other hand, the contact angle of the support bearing in the RV reducer is small, so only the radial support effect is considered in the modeling. Considering the impact of the above two aspects, in order to express the radial support function of the two sets of bearings, this paper doubles the total contact stiffness between the tapered roller and the inner/outer raceways and converts it to the radial direction. Therefore, the equivalent radial contact stiffness between the tapered roller and the inner/outer raceways K_r can be expressed as:

$$K_r = 2K_h \cos \alpha_e \quad (20)$$

3.2. Transmission pair contact modeling

3.2.1. Contact modeling of planetary gear pair

The contact modeling method for the involute planetary gear pair is shown in Fig. 6. The Coordinate system $x_k o_k y_k$ is the local coordinate system of the input gear shaft, and the coordinate systems $x_{d1} o_{d1} y_{d1}$ and $x_{d2} o_{d2} y_{d2}$ are the local coordinate systems of two sets of planetary gear and crank shaft assemblies. The pitch circle radius of the input gear shaft and planetary gear are R_k and R_d , respectively. In the model, the contact between gear teeth can be simulated by equivalent springs [51]. The symbol K_m represents the meshing stiffness of the gear pair, C_m represents the meshing damping, and b_t represents the dynamic side clearance. Due to the elastic influence of the supporting bearings on the planetary gear and the crankshaft assembly, there is radial runout at the center of the planetary gear. Therefore, the center distance and tooth side clearance between the input gear shaft and the planetary gear exhibit time-varying characteristics during meshing. In the model, the ideal center distance vectors between the input gear shaft and the two planetary gears are denoted by $\mathbf{a}_{kd}^1$ and $\mathbf{a}_{kd}^2$, while the actual center distance vectors between the input gear shaft and the two planetary gears are denoted by $\mathbf{a}_{kd}^{1'}$ and $\mathbf{a}_{kd}^{2'}$. The radial runout position vectors of the two planetary gear centers are denoted by $\mathbf{e}_{kd}^1$ and $\mathbf{e}_{kd}^2$.

To ensure lubrication and prevent tooth jamming caused by gear tooth deformation, a certain amount of initial side clearance is required for gear pairs. Based on the geometric properties of involute curves, when initial side clearance b_0 exists, the dynamic side clearance of gear pairs can be calculated as follows:

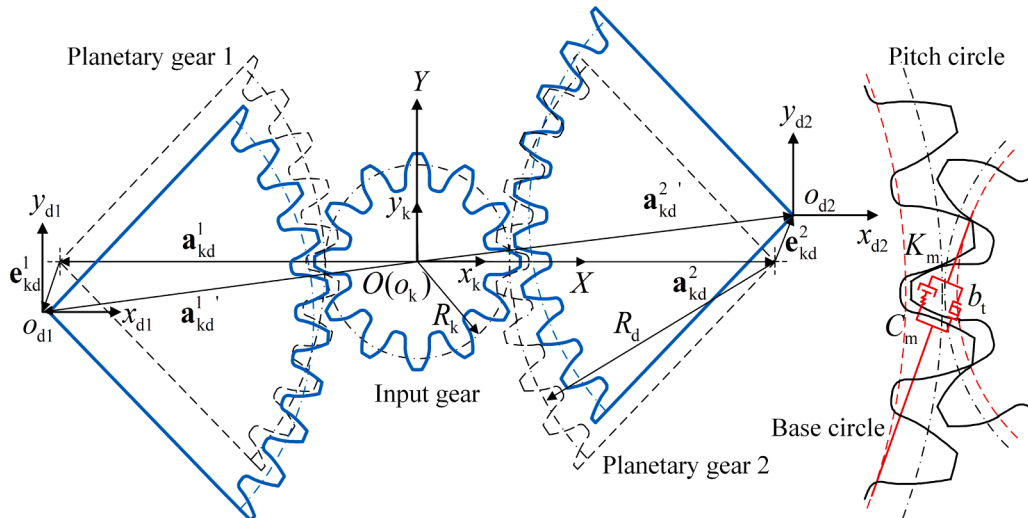


Fig. 6. Planetary gear pair contact modeling.

$$b_t = \left(\frac{2\pi R'_k}{Z_k} - s'_k - s'_d \right) \cos \alpha' + b_0 \quad (21)$$

where R'_k is the actual pitch circle radius of the input gear shaft, Z_k is the number of input gear shaft teeth, s'_k and s'_d are the pitch circle tooth thicknesses of the input gear shaft and planetary gear, respectively, α' is the actual meshing angle, α' can be calculated by the following equation:

$$\alpha' = \arccos\left(\frac{a}{a'} \cos(\alpha)\right) \quad (22)$$

where a is the ideal center distance, a' is the actual center distance, α is the ideal meshing angle.

According to the above two equations, the dynamic tooth side clearance of the gear pair can be expressed as:

$$b_t = 2a \cos(\alpha) [\text{inv}(\alpha') - \text{inv}(\alpha)] + b_0 \quad (23)$$

When considering the radial runout of the planetary gear center, the relative displacement between the input gear shaft and the planetary gear along the normal direction of the meshing line can be expressed as:

$$g_t = (R'_k \theta_k - R'_d \theta_d) \cos \alpha' + e_{kd}^x \cos \alpha' + e_{kd}^y \sin \alpha' \quad (24)$$

where R'_d is the actual pitch radius of the planetary gear, e_{kd}^x and e_{kd}^y are the runout amounts of the planetary gear in the coordinate axis direction, respectively, θ_k and θ_d are the rotation angles of the input gear shaft and planetary gear, respectively.

Introducing the tooth side clearance function to determine the meshing state of the gear pair:

$$f_t = \begin{cases} g_t & g_t \leq 0 \\ 0 & -b_t < g_t < 0 \\ |g_t + b_t| & g_t \leq -b_t \end{cases} \quad (25)$$

When $g_t \geq 0$, the gear pair is in a normal state of tooth surface contact, when $-b_t < g_t < 0$, the gear pair is in a disengaged state where the gear teeth are separated from each other, and when $g_t \leq -b_t$, the gear pair is in an abnormal state of tooth back contact.

The contact force of the gear pair can be calculated by the following equation:

$$F_{kd} = K_m (f_t)^{10/9} + C_m \dot{g}_t \quad (26)$$

where $\dot{g}_t$ is the relative velocity in the direction of the meshing line, C_m is the meshing damping, which varies with the tooth side clearance. It can be calculated as follows:

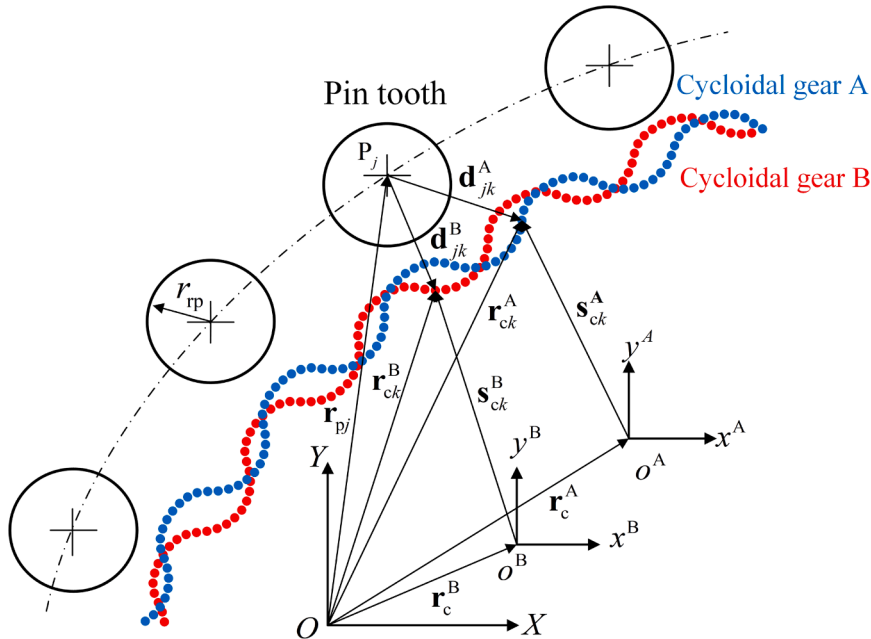


Fig. 7. Cycloid-pin pair contact modeling.

$$C_m = \begin{cases} 0 & f_i < l_0 \\ C \frac{f_i - l_0}{l_1 - l_0} & l_0 \leq f_i \leq l_1 \\ C & f_i > l_1 \end{cases} \quad (27)$$

where C is the given value of the meshing damping, the damping value in the contact modeling can be selected based on experiments conducted by Mitsuya [52], contact damping in subsequent contact modeling also refers to this literature, l_0 and l_1 are the critical values of contact depth of the gear pair, $l_1 > l_0$.

3.2.2. Contact modeling of the cycloid-pin pair

Contact modeling of the cycloid-pin pair first substitutes the continuous cycloid gear profile with a finite number of discrete points, and then determines whether the pin teeth are in contact with the cycloid gear by calculating the distance between the center of the pin tooth and the discrete points on the cycloid gear profile. As shown in Fig. 7, coordinate systems $x^A o^A y^A$ and $x^B o^B y^B$ are the local coordinate systems of cycloid gears A and B, respectively. The position vector of any discrete point on the cycloid gear profile in the generalized coordinate system XOY can be described as:

$$\mathbf{r}_{ck}^Z = \mathbf{r}_c^Z + \mathbf{A}_c^Z \mathbf{s}_{ck}^Z (Z = A, B) \quad (28)$$

where $\mathbf{r}_c^Z$ is the position vector of the origin of the local coordinate system of the cycloid gear in the generalized coordinate system, $\mathbf{s}_{ck}^Z$ is the position vector of the discrete points on the cycloid gear profile in the local coordinate system of the cycloid gear, $\mathbf{A}_c^Z$ is the coordinate transformation matrix of cycloid gear in the generalized coordinate system, k is the serial number of the discrete points on the cycloid gear profile.

The relative position vector between the discrete points on the cycloid gear profile and the center of the pin tooth can be described as:

$$\mathbf{d}_{jk}^Z = \mathbf{r}_{ck}^Z - \mathbf{r}_{pj} \quad (29)$$

where $\mathbf{r}_{pj}$ is the position vector of the needle tooth center in the generalized coordinate system, j is the serial number of the pin tooth.

When the cycloid gear and the pin tooth are in contact, the relationship between the distance from the discrete point on the cycloid gear profile to the center of the pin tooth and the radius of the pin tooth must satisfy the following condition:

$$\|\mathbf{d}_{jk}^Z\| - r_{tp} \leq 0 \quad (30)$$

Take cycloid gear A as an example. As shown in Fig. 8, at a specific contact position, multiple discrete points on the cycloid gear profile come into contact with the pin tooth. In the contact analysis of the cycloid-pin pair, the discrete point with the biggest contact depth is taken as the point of force application between the cycloid gear and the pin tooth. By calculating the distance between the center of each pin tooth and the discrete points on the cycloid gear profile, the maximum contact depth between the cycloid gear and the pin tooth can be screened out. The maximum contact depth can be described as:

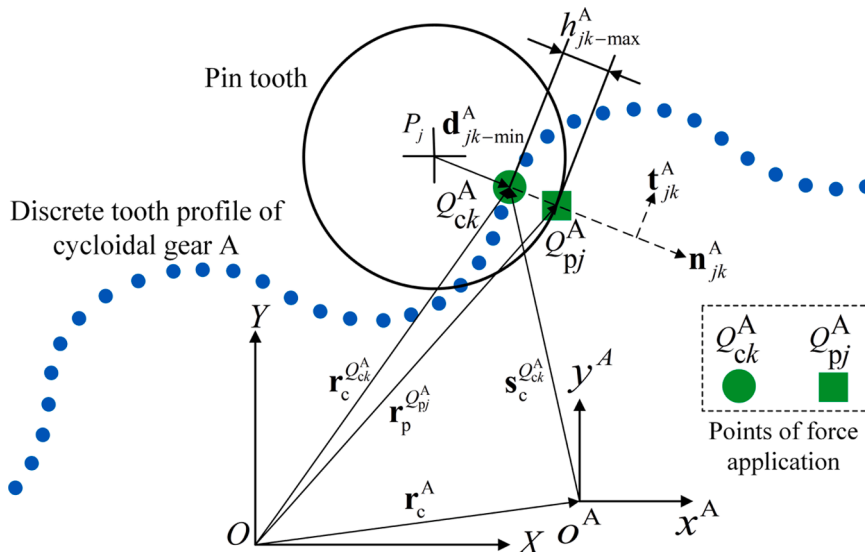


Fig. 8. Contact detection between the cycloid gear and the pin tooth.

$$h_{jk-\max}^A = \max \| |\mathbf{d}_{jk}^A| - r_{rp} | \quad (31)$$

Once the maximum contact depth has been screened out, the minimum relative position vector between the discrete points of the cycloid gear profile and the center of the pin tooth $\mathbf{d}_{jk-\min}^Z$ can be determined. The direction of the normal contact force can be represented by the following unit vector:

$$\mathbf{n}_{jk}^A = \frac{\mathbf{d}_{jk-\min}^A}{\| \mathbf{d}_{jk-\min}^A \|} \quad (32)$$

As shown in Fig. 8, the point Q_{ck}^A is the point of force application on cycloid gear A, and its relative position vector in the coordinate system XOY can be described as:

$$\mathbf{r}_c^{Q_{ck}^A} = \mathbf{r}_c^A + \mathbf{A}_c^A \mathbf{s}_{ck}^{Q_{ck}^A} \quad (33)$$

where $\mathbf{s}_{ck}^{Q_{ck}^A}$ is the relative position vector of the point of force application on the cycloid gear profile in the local coordinate system of the cycloid gear.

The relative velocity vector of the point of force application can be obtained by differentiating its relative position vector with respect to time:

$$\dot{\mathbf{r}}_c^{Q_{ck}^A} = \dot{\mathbf{r}}_c^A + \dot{\mathbf{A}}_c^A \mathbf{s}_{ck}^{Q_{ck}^A} \quad (34)$$

The direction of the tangential friction force is represented by the vector $\mathbf{t}_{jk}^A$, which can be obtained by rotating the vector $\mathbf{n}_{jk}^A$ counterclockwise by 90° . The relative normal velocity and relative tangential velocity between the tooth profile discrete point and the pin tooth can be expressed as:

$$V_{cp-n}^A = \left(\dot{\mathbf{r}}_c^{Q_{ck}^A} - \dot{\mathbf{r}}_p^{Q_{pj}^A} \right)^T \mathbf{n}_{jk}^A \quad (35)$$

$$V_{cp-t}^A = \left(\dot{\mathbf{r}}_c^{Q_{ck}^A} - \dot{\mathbf{r}}_p^{Q_{pj}^A} \right)^T \mathbf{t}_{jk}^A \quad (36)$$

In the contact modeling of the cycloid-pin pair, it is assumed that the position of the pin tooth is fixed and does not rotate, so the relative velocity vector $\dot{\mathbf{r}}_p^{Q_{pj}^A}$ at the point of force application Q_{pj}^A on the pin tooth is a zero vector.

The normal contact force between the discrete point on the cycloid gear profile and the pin tooth can be calculated using the following formula:

$$F_{cp-n}^A = K_{cp} \left(h_{jk-\max}^A \right)^{10/9} + c_c C_{cp} V_{cp-n}^A \quad (37)$$

where c_c is the adjustment coefficient for contact damping of the cycloid-pin pair, C_{cp} is the contact damping of the cycloid-pin pair.

The adjustment coefficient for contact damping serves to prevent the discontinuity of the damping force during the contact process and ensure the stability of the contact analysis. It can be calculated by the following formula:

$$c_c = \begin{cases} 0 & h_{jk-\max}^A < h_0 \\ \frac{h_{jk-\max}^A - h_0}{h_1 - h_0} & h_0 \leq h_{jk-\max}^A \leq h_1 \\ 1 & h_{jk-\max}^A > h_1 \end{cases} \quad (38)$$

where h_0 and h_1 are the critical values of contact depth of the cycloid-pin pair, $h_1 > h_0$.

The tangential friction force between the discrete point on the cycloid gear profile and the pin tooth can be calculated by the following equation:

$$F_{cp-t}^A = -\mu_f c_f F_{cp-n}^A V_{cp-t}^A / \| V_{cp-t}^A \| \quad (39)$$

where μ_f is the dynamic friction coefficient, c_f is the adjustment coefficient for the friction force.

The friction coefficient adjustment factor ensures the smooth application of friction during the contact process, thereby ensuring the stability of the contact analysis. It can be calculated by the following equation:

$$c_f = \begin{cases} 0 & V_{cp-t}^A < V_0 \\ \frac{V_{cp-t}^A - V_0}{V_1 - V_0} & V_0 \leq V_{cp-t}^A \leq V_1 \\ 1 & V_{cp-t}^A > V_1 \end{cases} \quad (40)$$

where V_0 and V_1 are the critical values of relative velocity, $V_1 > V_0$.

3.2.3. Contact modeling of the swivel arm bearing and support bearing

The inner and outer raceways of the swivel arm bearing of the RV reducer are the eccentric circle of the crankshaft and the bearing hole on the cycloid gear, respectively. The contact model of the swivel arm bearing is shown in Fig. 9. The coordinate system $x_d o_d y_d$ is the local coordinate system of the crankshaft. When the swivel arm bearing is under load, the geometric center of the inner raceway o_i and the geometric center of the outer raceway o_o will shift and no longer coincide. The relative position vector of the geometric centers of the inner and outer raceways in the generalized coordinate system XOY can be expressed as:

$$\mathbf{r}_i = \mathbf{r}_d + \mathbf{A}_d \mathbf{s}_d \quad (41)$$

$$\mathbf{r}_o = \mathbf{r}_c^A + \mathbf{A}_c^A \mathbf{s}_{co} \quad (42)$$

where $\mathbf{r}_d$ is the position vector of the origin of the crankshaft local coordinate system in the generalized coordinate system, $\mathbf{A}_d$ is the coordinate transformation matrix of the crankshaft in a generalized coordinate system, $\mathbf{s}_d$ is the position vector corresponding to the eccentricity of the crankshaft, $\mathbf{s}_{co}$ is the position vector of the center of the outer raceway of the bearing in the local coordinate system of cycloid gear A.

The relative eccentricity position vector and eccentricity velocity vector between the geometric centers of the inner and outer raceways of the swivel arm bearing can be expressed as follows:

$$\mathbf{e}_{io} = \mathbf{r}_i - \mathbf{r}_o \quad (43)$$

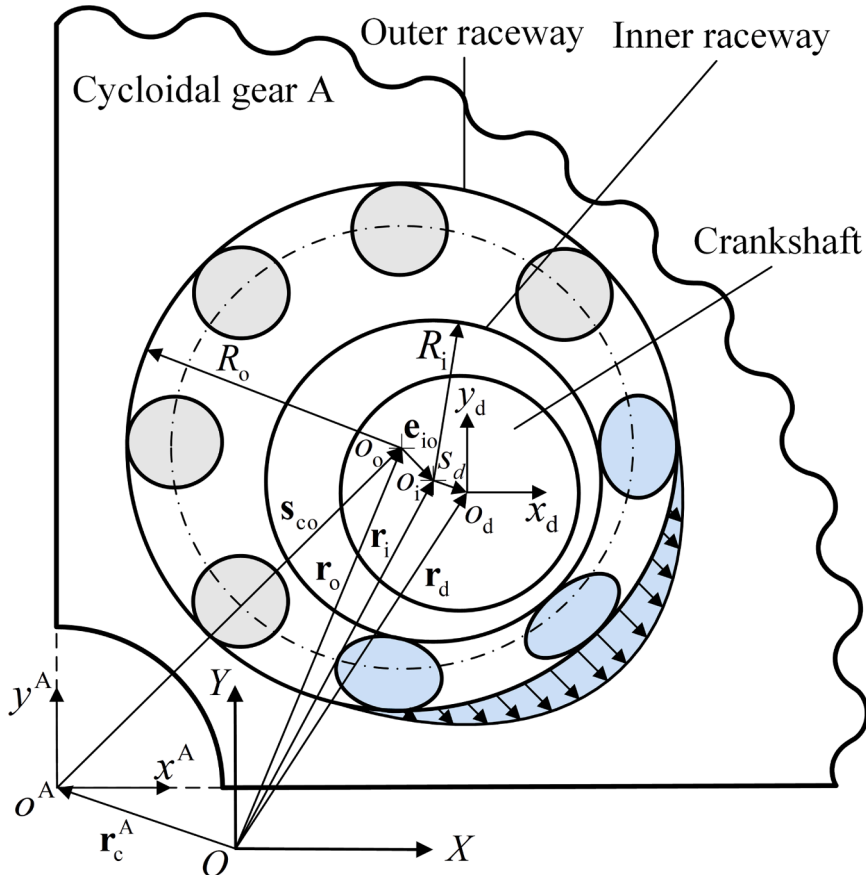


Fig. 9. Swivel bearing contact modeling.

$$\mathbf{v}_{io} = \dot{\mathbf{r}}_i - \dot{\mathbf{r}}_o \quad (44)$$

The eccentricity between the geometric centers of the inner and outer raceways of the swivel arm bearing can be calculated as follows:

$$\|\mathbf{e}_{io}\| = \sqrt{(\mathbf{e}_{io})^T \mathbf{e}_{io}} \quad (45)$$

Assuming that the rollers are evenly distributed between the inner and outer raceways of the swivel arm bearing and have the same rotational speed, the angular velocity of the bearing rollers can be expressed as:

$$\omega_g = \frac{\omega_i R_i + \omega_o R_o}{R_i + R_o} \quad (46)$$

where R_i and R_o are the radius of the inner and outer raceways, respectively, ω_i and ω_o are the angular velocities of the inner and outer raceways, respectively.

During the rotation of the swivel arm bearing, the angle positions of each cylindrical roller can be calculated as follows:

$$\varphi_g^u = \varphi_0^u + \omega_g t \quad (47)$$

where φ_g^u is the initial position angle of each cylindrical roller, u is the serial number of the cylindrical roller.

During rotation, the radial deformation of the cylindrical roller at different angular positions can be expressed as:

$$\delta_g^u = e_{io-x} \cos \varphi_g^u + e_{io-y} \sin \varphi_g^u \quad (48)$$

where e_{io-x} and e_{io-y} is the displacement component of the eccentricity vector $\mathbf{e}_{io}$ in the direction of the coordinate axis of the coordinate system XOY .

The relative eccentric velocity of the cylindrical roller at different angular positions can be expressed as:

$$\mathbf{v}_g^u = v_{io-x} \cos \varphi_g^u + v_{io-y} \sin \varphi_g^u \quad (49)$$

where v_{io-x} and v_{io-y} is the velocity component of the eccentricity velocity vector $\mathbf{v}_{io}$ in the direction of the coordinate axis of the coordinate system.

The normal contact force between each roller and the raceways can be described as:

$$F_g^u = K_g \left(\delta_g^u \right)^{10/9} + C_g v_g^u \quad (50)$$

where C_g is the contact damping between the cylindrical roller and the raceways.

The contact modeling method for the support bearing is similar to that of the swivel arm bearing, with the exception that the support bearing needs to consider the effect of bearing preload. The contact model of the support bearing can be found in literature [3].

Table 1
Geometric and structural design parameters.

Components	Parameters	Values
Input gear shaft	Number of input gear shaft teeth Z_k	12
	Pressure angle α (°)	20
Planetary gear	Number of planetary gear teeth Z_d	36
	Pressure angle α (°)	20
Cycloid gear	Number of cycloid gear teeth, Z_c	39
Pin gear	Number of pins Z_p	40
	Pin tooth radius r_{tp} (mm)	3
	Radius of pin tooth distribution circle r_p (mm)	66
Crankshaft	Crankshaft eccentricity s_d (mm)	1.3
Swivel arm bearing (KZK26×36×12)	Number of rollers	14
	Inner diameter (mm)	26
	Outer diameter (mm)	36
	Height (mm)	12
Support bearing (30,202)	Number of rollers	14
	Contact angle α_c (°)	15°
	Inner diameter (mm)	15
	Outer diameter (mm)	35
	Height (mm)	13.5

4. Simulation analysis of the RV reducer hysteresis curve and experimental verification

4.1. Parameters for the dynamic analysis

In this paper, the RV40E-121 reducer is taken as the research object to conduct the simulation analysis and experimental research on the hysteresis curve. The geometric and structural design parameters of the main transmission components are shown in Table 1.

Mass and moment of inertia parameters of the main transmission components of the RV40E-121 reducer are as shown in Table 2.

Based on engineering experience, the machining accuracy of the cycloid gear, crankshaft, and pin teeth has a significant impact on the transmission performance of the RV reducer. Therefore, the influence of geometric errors in components should be considered in modeling. The geometric errors modeling method of the RV reducer can be found in the literature [3]. This method uses the RAND function to assign values to each geometric error within the tolerance range. We utilized the coordinate measuring machine (CMM) to inspect the cycloid tooth profile, crankshaft eccentric circle, eccentricity length, bearing hole, swivel arm bearing roller, and pin tooth. The design of geometric error ranges refers to the accuracy inspection results obtained from the CMM. The geometric errors design ranges of the main transmission components of the reducer is listed in Table 3. In addition to the errors listed in Table 3, the diameter error of each roller of the swivel arm bearing is set to $-2.5\ \mu\text{m}$. The initial tooth side clearance of the planetary gear pair is set to $2\ \mu\text{m}$.

Taking the cycloid gear as an example, its actual tooth profile includes various errors such as pitch error, radial runout of the tooth circle, and surface waviness, which makes the precise characterization of cycloid tooth profile significantly difficult. In addition, in order to ensure good oil film lubrication characteristics of the cycloid-pin pair, a reasonable tooth side clearance is required between the cycloid tooth profile and the pin teeth. In this paper, the “negative isometric modification + negative offset modification” method [38] is used to simplify the tooth profile error and tooth side clearance of the cycloid-pin pair. Based on the CMM inspection result, by comparing the actual tooth profile curve with the ideal tooth profile curve, the maximum outer deviation and minimum inner deviation of the actual tooth profile relative to the ideal tooth profile are extracted. In this paper, the isometric modification and offset modification amounts are set to $-5\ \mu\text{m}$ and $-15\ \mu\text{m}$, respectively. The tooth profile deviation after modification is between the maximum external deviation and the minimum internal deviation.

Dynamic simulation parameters are shown in Table 4. The process for solving the dynamics model of RV reducer is shown in Fig. 10.

4.2. Hysteresis curve simulation analysis of RV reducer

In the dynamic model of the RV reducer, the three degrees of freedom of the input gear shaft are constrained to fix the input gear shaft. The loading and unloading of torque are carried out step by step on the output disc of the reducer according to the method described in Section 2. As shown in Fig. 11, the control strategy of load torque loading and unloading at the output end of the reducer contains five stages: forward loading (stage 1), reverse unloading (stage 2), reverse loading (stage 3), forward unloading (stage 4), and forward loading (stage 5). At each stage, the loading or unloading of torque is divided into 30 steps. The hysteresis curve obtained through dynamic simulation at the initial position of the RV reducer is shown in Fig. 12. In the figure, the red dotted line corresponds to the result of the reducer torsion angle change obtained from the first stage of torque loading. The reducer angle hysteresis change obtained in this stage cannot be used for the evaluation of the torsional rigidity and lost motion, so the results of the first stage of torque loading are no longer given in the subsequent hysteresis curve analysis. The closed hysteresis curve of the reducer corresponds to the torque loading and unloading process in stages two to five.

The torsional rigidity is calculated using the torque and corresponding torsional angle in the range of $(2/3)T_r$ to T_r . Based on the hysteresis curve obtained from the simulation, the torsional rigidity of the reducer at the initial position can be calculated to be $131.98\ \text{N}\cdot\text{m}/\text{arcmin}$, and the lost motion is $0.364\ \text{arcmin}$.

Generally, the torsional rigidity and lost motion characteristics of the RV reducer are influenced by the relative meshing position or contact state of the internal transmission components. To explore the influence of the relative position changes of internal transmission components on the torsional rigidity and lost motion characteristics of the reducer, it is necessary to drive the input gear shaft to rotate the output disc by a certain angle, then fix the input gear shaft, and finally apply torque to the output disc in steps in the dynamic model. Within the range of one full rotation of the output disc, a hysteresis curve simulation is conducted for every 9° rotation of the output disc, and the hysteresis curves of the reducer at 40 working positions can be obtained. The torsional rigidity and lost motion variation curves at different operating positions of the reducer are shown in Figs. 13 and 14, respectively.

As shown in Fig. 13, the torsional rigidity of the reducer fluctuates between $125.01\ \text{N}\cdot\text{m}/\text{arcmin}$ and $143.57\ \text{N}\cdot\text{m}/\text{arcmin}$, with an arithmetic mean of $133.04\ \text{N}\cdot\text{m}/\text{arcmin}$. As shown in Fig. 14, the lost motion of the reducer fluctuates between $0.321\ \text{arcmin}$ and $0.412\ \text{arcmin}$, with an arithmetic mean of $0.355\ \text{arcmin}$.

Table 2
Mass and moment of inertia parameters.

Components	Mass m (kg)	Moment of inertia I ($\text{kg}\cdot\text{mm}^2$)
Input gear shaft	0.458	77
Planetary gear and crankshaft assembly	0.258	49
Cycloid gear	0.467	1160
Output assembly	3.45	7740

Table 3
The ranges of geometric errors.

Components	Error terms	Lower deviation	Upper deviation
Crankshaft	Eccentric circle radius error (μm)	-3	0
	eccentric circle distribution angle error ($^{\circ}$)	-60	+60
	Eccentricity length error (μm)	-2	+2
Cycloid gear	Cycloid gear bearing hole distribution circle radius error (μm)	-3	+3
	Cycloid gear bearing hole radius error (μm)	0	+3
	Cycloid gear bearing hole distribution angle error ($^{\circ}$)	-60	+60
Pin gear	Pin tooth distribution circle radius error (μm)	0	3
	Pin tooth radius error (μm)	-3	0
	Pin tooth distribution angle error ($^{\circ}$)	-30	+30
Output assembly	Output disc bearing hole distribution circle radius error (μm)	-3	+3
	Output disc bearing hole distribution angle error ($^{\circ}$)	-60	+60

Table 4
Dynamic simulation parameters.

Dynamic simulation parameters	Values
Rated torque T_r (N·m)	412
Gear pair meshing stiffness K_m (N/m)	$(0.8-1.48) \times 10^8$
cycloid-pin pair contact stiffness, K_{cp} (N/m)	$(0.25-15.5) \times 10^8$
Swivel arm bearing roller-raceway contact stiffness K_g (N/m)	2.38×10^8
Supporting arm bearing roller-raceway contact stiffness K_r (N/m)	4.01×10^8
Planetary gear pair, cycloid-pin pair, bearing roller-raceway contact damping C, C_{cp}, C_g, C_h (N·s/m)	500
Support bearing axial preload displacement (mm)	0.01
Gravitational acceleration g (m/s^2)	9.8
Numerical calculation methods	ode45
Integral solution time step (ms)	0.002

4.3. Experiment verification

To verify the accuracy of the proposed dynamic model, this paper conducted torsional rigidity and lost motion experiment on the RV40E-121 reducer. The torsional rigidity and lost motion test bench is shown in Fig. 15. The test bench mainly consists of a drive motor, a load motor, a planetary reducer, angle encoders at the drive end and load end, torque sensors at the drive end and load end, the tested RV reducer, and a pneumatic brake. The brands and performance parameters of the above equipment are listed in Table 5. During the test, when generating the hysteresis curve, the pneumatic brake was used to lock the input gear shaft of the reducer. The servo motor at the load end applied torque to the output disc of the reducer in stages, and the angle encoder and torque sensor were used to synchronously record the torsional angle and torque at the output end. The torque loading method at the output end is consistent with the process described in Section 2. During the test, the loading speed of the load motor is 0.3 r/min. The planetary reducer serves to enhance the output torque of the load motor.

Through the experiment, the hysteresis curve of the RV reducer, as shown in Fig. 19, can be plotted. Based on the hysteresis curve shown in Fig. 16, the torsional rigidity of the reducer at the initial position can be evaluated as 116.92 N·m/arcmin, and the lost motion is 0.517 arcmin.

After completing the hysteresis curve test at a certain working position of the reducer and conducting the hysteresis curve test at the next working position, it is necessary to release the pneumatic brake and drive the input gear shaft of the reducer through the drive motor to rotate the output disc by a certain angle. Following this, it needs to use the pneumatic brake to re-lock the input gear shaft of the reducer, and apply torque to the output disc of the reducer in stages through the load motor again.

The hysteresis curves of the RV reducer at 40 working positions were also tested through experiments. Hysteresis curve testing is performed once every 9° of output disc rotation. However, the RV reducer is a closed-type transmission unit, which makes it difficult to determine the relative meshing position or contact state of the internal transmission components. In the simulation, the meshing positions of planetary gear pair and cycloid-pin pair, as well as the relative contact state between rollers and raceways in multiple sets of bearings, are difficult to remain consistent with the experiment. Therefore, the comparison between the experiment results and the simulation results should avoid comparing the torsional rigidity and lost motion at each working position separately. When comparing the hysteresis curves, the analysis should only focus on the shape of the hysteresis curves. When conducting data analysis, the analysis should only focus on the statistical characteristics of the torsional rigidity and lost motion values of the 40 working positions, such as the average, minimum, and maximum values. Fig. 17 illustrates the simulated and experimental hysteresis curves of the RV reducer at 8 working positions. It can be observed that, compared to the experimental hysteresis curve, the simulated hysteresis curves exhibit a good simulation effect in terms of shape.

Similarly, the torsional rigidity is calculated using the torque and corresponding torsional angle in the range of $[(2/3)T_r, T_r]$. The torsional rigidity and lost motion variation curves of the RV reducer at different working positions obtained from the test are shown in Figs. 18 and 19, respectively. As shown in Fig. 21, the torsional rigidity obtained through the experiment fluctuates between 110.82 N·m/arcmin and 131.46 N·m/arcmin, with an arithmetic mean of 123.10 N·m/arcmin. As shown in Fig. 19, the lost motion obtained

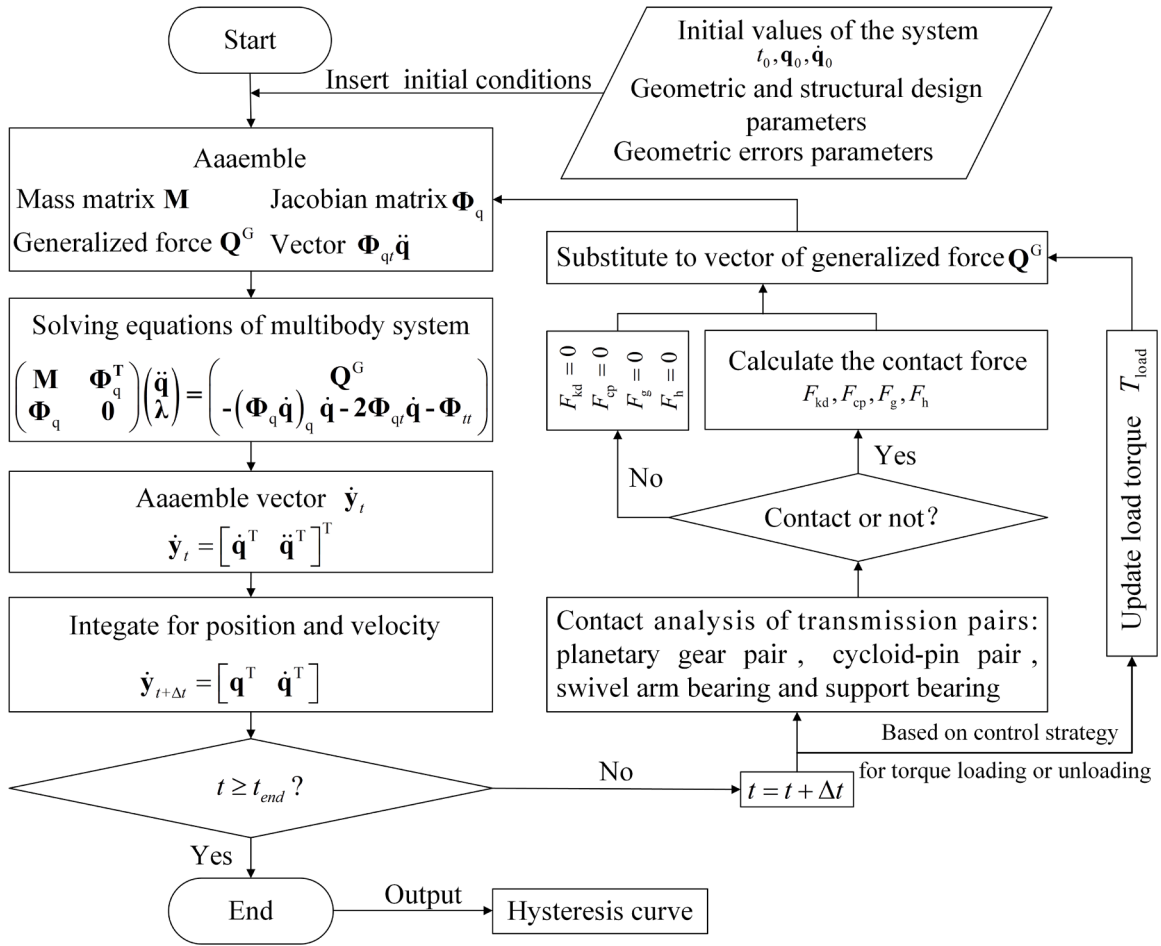


Fig. 10. Flowchart for solving the dynamics model of RV reducer.

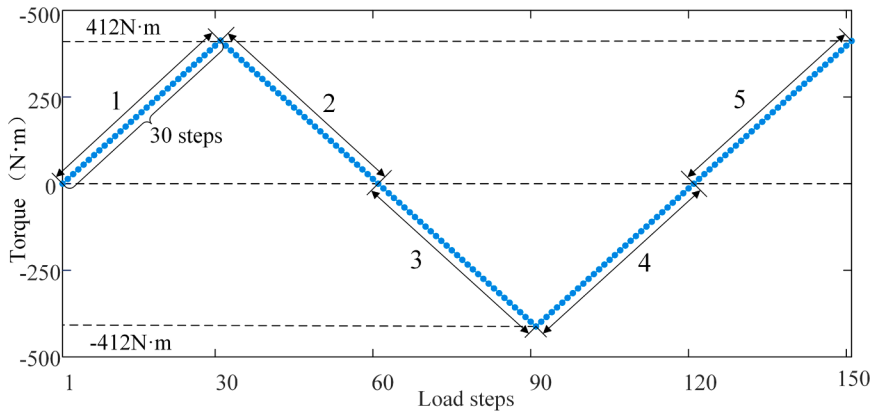


Fig. 11. Torque loading and unloading control strategy at the output end of RV reducer.

through the experiment fluctuates between 0.317 " and 0.667 arcmin, with an arithmetic mean of 0.455 arcmin.

Comparing the simulation results with the experimental results, the average value of the simulated torsional rigidity differs from the average value of the experimental results by 8.07 %, the maximum value differs by 9.25 %, and the minimum value differs by 12.8 %. The average value of the simulated lost motion differs from the experimental results by 21.98 %, the maximum value differs by 38.23 %, and the minimum value differs by 1.26 %. Compared to experimental results, the simulation results exhibit higher torsional rigidity and lower lost motion. This is mainly due to the simplification of the geometric errors model. Assembly errors of the reducer,

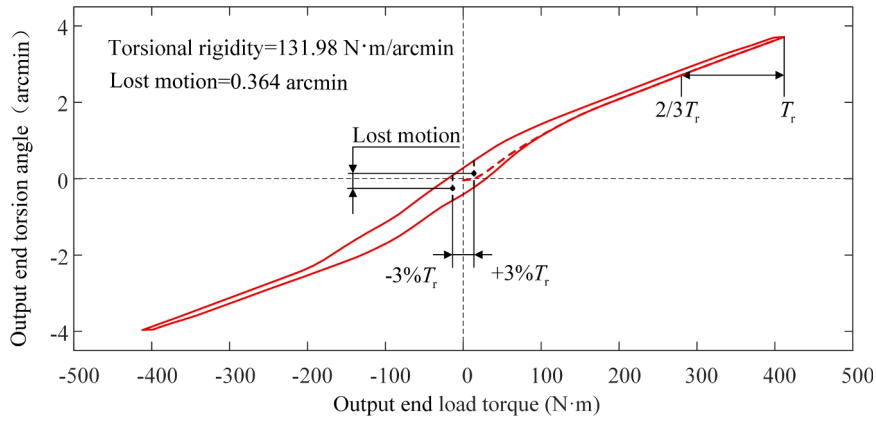


Fig. 12. Hysteresis curve of simulation.

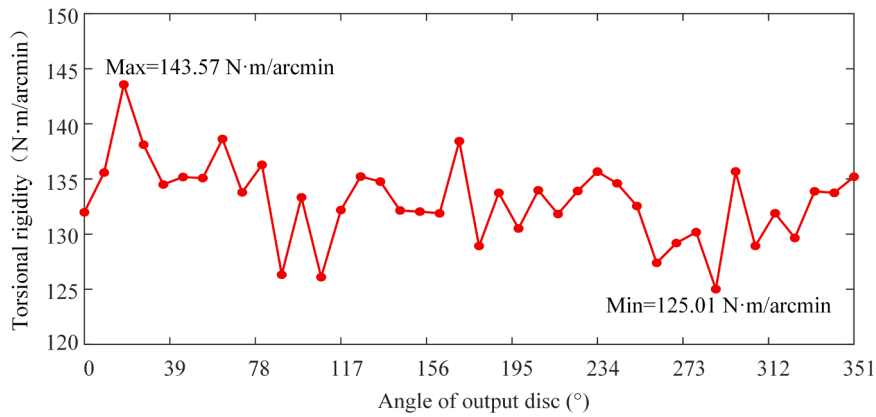


Fig. 13. Torsional rigidity variation curve (simulation).

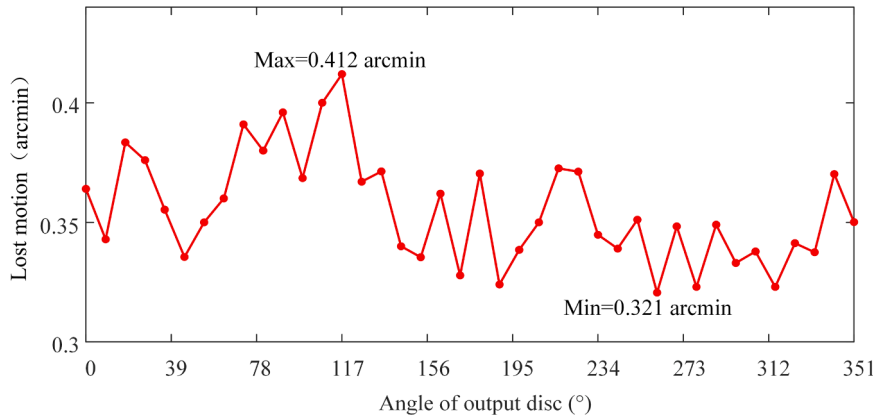


Fig. 14. Lost motion variation curve (simulation).

geometric and positional errors of its components, and tooth profile errors of the cycloid gear are difficult to fully and accurately express in our contact dynamic model. In the theoretical model, the simplification of geometric error model will ignore a part of the clearance effect, idealizing the contact state of the transmission pairs. The simplification can lead to the number of meshing teeth of the cycloid-pin pair and the number of loaded bearing rollers being more than the actual condition, which in turn results in a larger torsional rigidity and a smaller lost motion in the simulation results. In addition, minor mechanical clearances exist at various connection parts of the test bench. The mechanical clearances will cause the experimental results to include additional torsional deformation and lag angle, resulting in a lower measured torsional rigidity and higher lost motion for the reducer.

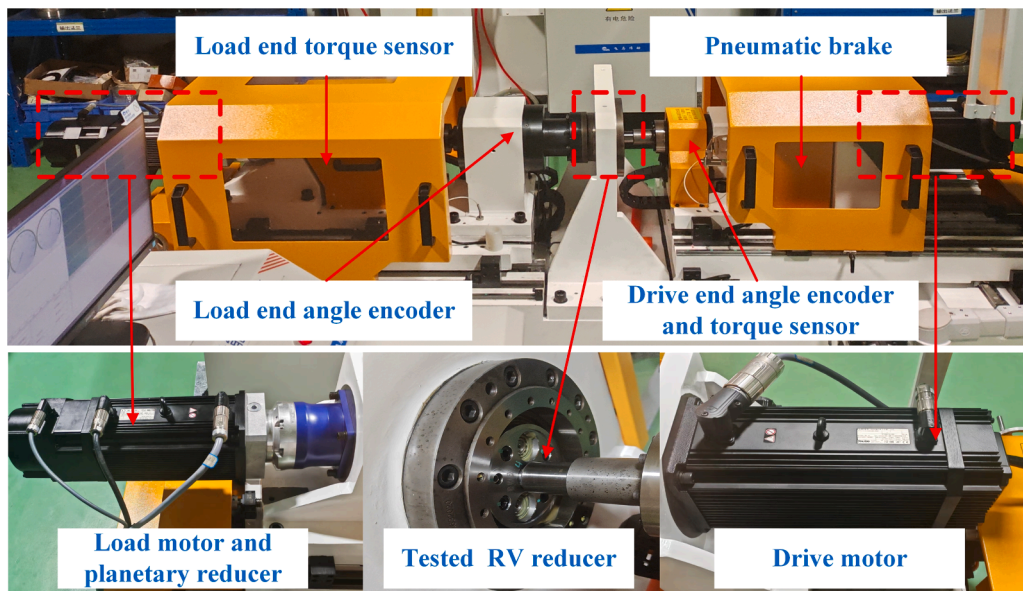


Fig. 15. Torsional rigidity and lost motion test bench.

Table 5

Parameters of test bench equipment.

Equipment	Brand	Performance parameters	
Drive motor	Lenze	Power	5.2 KW
Load motor	Lenze	Power	6.9 KW
Drive end angle encoder	Renishaw	Precision	$\pm 1.5''$
Load end angle encoder	Renishaw	Precision	$\pm 1.5''$
Drive end torque sensor	Kistler	Precision	$\pm 0.1\%$ F.S
		Range	10 N·m
Load end torque sensor	Kistler	Precision	$\pm 0.05\%$ F.S
		Range	500 N·m
Planetary reducer	Lenze	Reduction ratio	25
Pneumatic brake	J&Gong	Maximum pressure	0.7 MPa

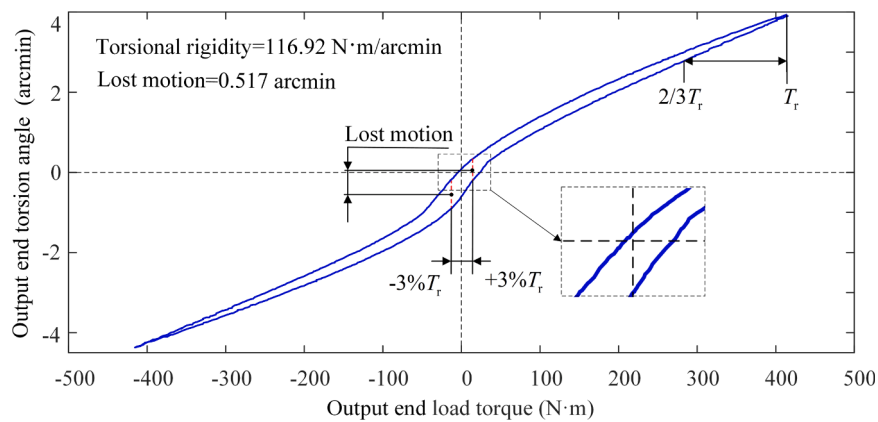


Fig. 16. Hysteresis curve of experiment.

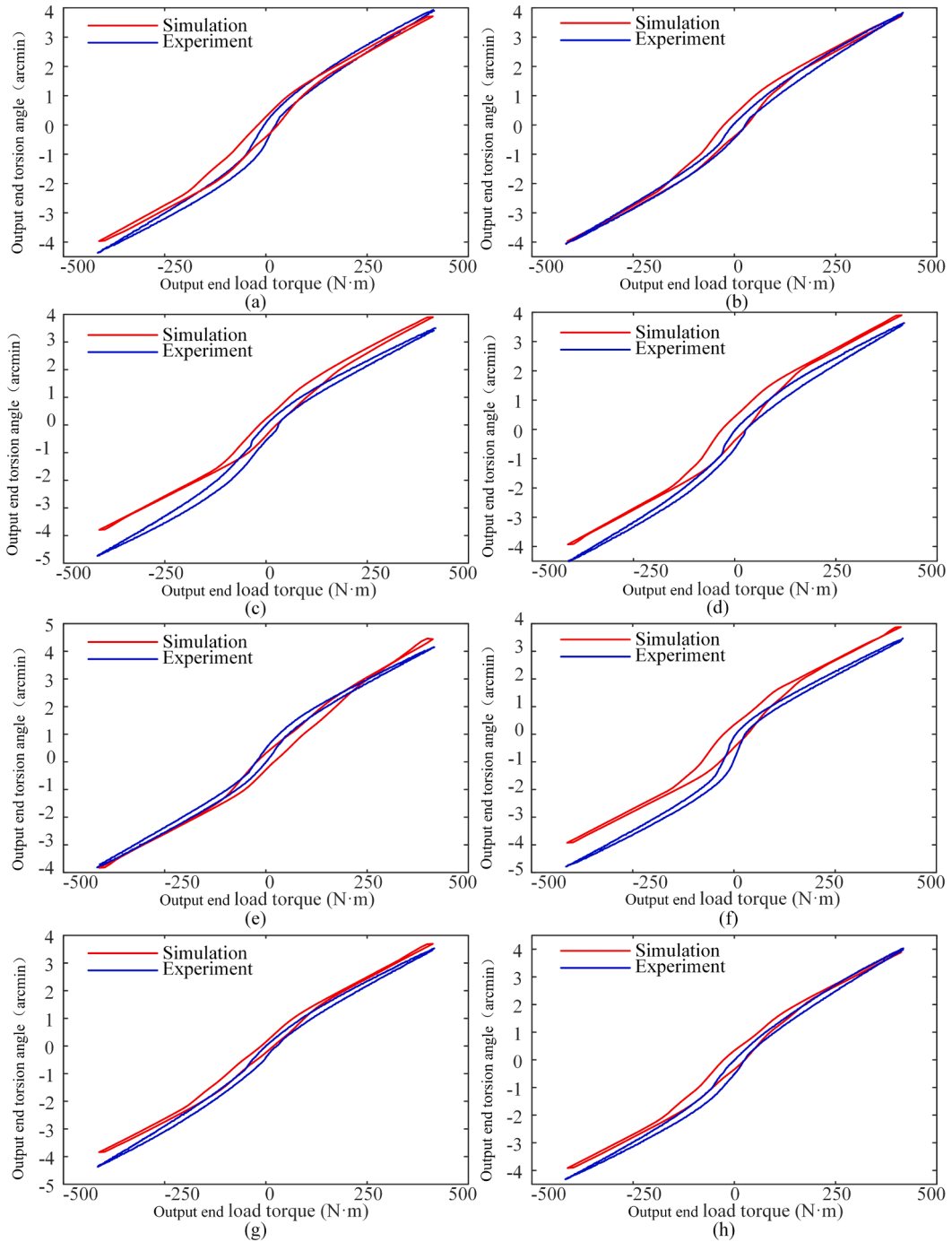


Fig. 17. Comparison of simulated hysteresis curves and experimental hysteresis curves : (a)0°, (b)45°, (c)90°, (d)135°, (e)180°, (f)225°, (g)270°, (h)315°.

5. Parametric analysis of torsional rigidity and lost motion of the RV reducer

5.1. Effect of contact stiffness on torsional rigidity and lost motion

The RV reducer dynamic model includes four types of contact stiffness: involute gear pair meshing stiffness, cycloid-pin pair meshing stiffness, roller-raceway contact stiffness of the swivel arm bearing, and roller-raceway contact stiffness of the support bearing. This paper investigates the effect of contact stiffness on torsional rigidity by adding identical increments to each contact

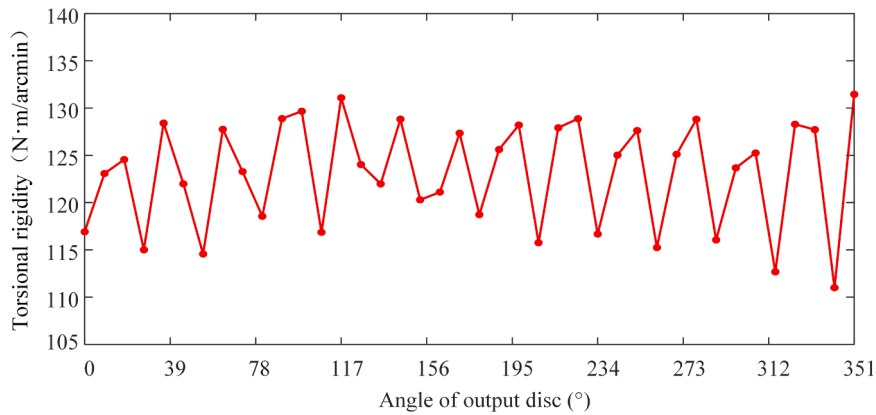


Fig. 18. Torsional rigidity variation curve (experiment).

stiffness value or stiffness range in Table 4. Fig. 20 illustrates the effects of changes in contact stiffness of each part on the torsional rigidity of the RV reducer. It can be seen that the roller-raceway contact stiffness of the support bearing has the most significant effect on the torsional rigidity of the RV reducer, followed by the roller-raceway contact stiffness of the swivel arm bearing. The contact stiffness of the cycloid-pin pair and the meshing stiffness of the involute gear pair have a minor impact on the torsional rigidity of the RV reducer, with the involute gear meshing stiffness having the least effect. The supporting bearing is installed on the output disc of the reducer and directly bears the load torque acting on the output disc. Moreover, under the preload effect, all the rollers of the supporting bearing are loaded, so its contact stiffness has the greatest impact on the torsional rigidity of the reducer. The reason why the meshing stiffness of the gear pair has the least impact on the torsional rigidity of the RV reducer is that the planetary gear pair is located in the high-speed stage of the reducer far from the output end.

Generally, the contact stiffness can be increased by increasing the tooth surface contact width of the involute gear, the tooth surface contact width of the cycloid gear, and the length of the bearing roller, thereby achieving the goal of improving the torsional rigidity of the reducer. Figs. 21 and 22 show the changes in the hysteresis curve and torsional rigidity of the reducer when the length of the bearing rollers and the tooth surface contact width of the cycloid gear and involute gear are increased, respectively. As shown in Fig. 21 (a) and (b), increasing the roller length of the support bearing and swivel arm bearing causes a significant change in the inclination of the hysteresis curve. As shown in Fig. 21(c), increasing the tooth surface contact width of the cycloid gear causes a slight change in the inclination of the hysteresis curve, while increasing the tooth surface contact width of the involute gear causes virtually no change in the inclination of the hysteresis curve, as shown in Fig. 21(d). The inclination of the hysteresis curve reflects the magnitude of torsional rigidity. As shown in Fig. 25, the length of the bearing roller has a significant impact on torsional rigidity. Furthermore, as shown in Fig. 22, increasing the length of the bearing rollers, particularly the support bearing rollers, can significantly enhance the torsional rigidity of the RV reducer.

It can be seen from Fig. 21(b) that in the small torque range, when the length of the swivel arm bearing roller increases, the distance between the upper and lower lines of the hysteresis curve decreases significantly. As shown in Fig. 21(a) and (c), within the small torque range, increasing the length of the support bearing roller and the contact width of the cycloid gear tooth surface results in a slight change in the distance between the upper and lower lines of the hysteresis curve. As shown in Fig. 21(d), in the small torque range, when the tooth surface contact width of the involute gear increases, the distance between the upper and lower lines of the

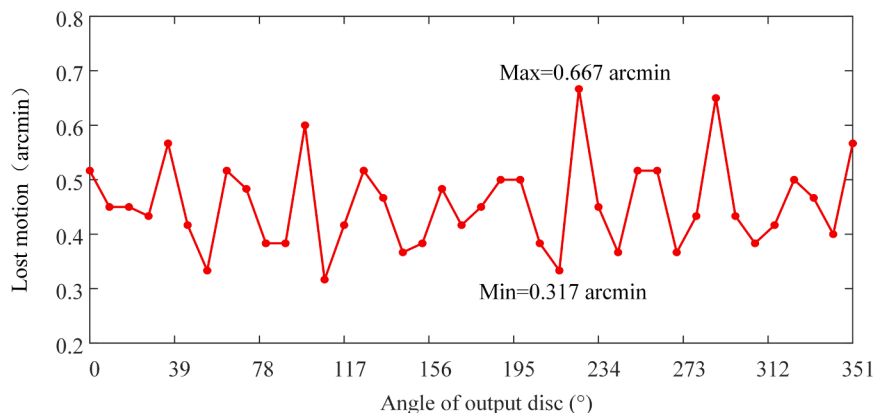


Fig. 19. Lost motion variation curve (experiment).

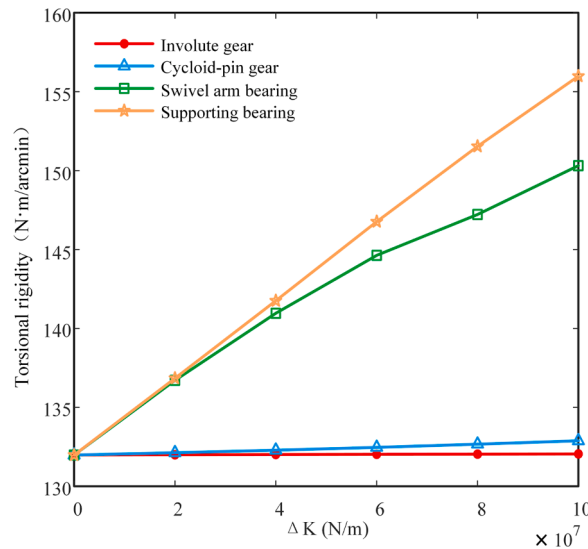


Fig. 20. Effect of contact stiffness on torsional rigidity.

hysteresis curve remains almost unchanged. The lost motion of an RV reducer is evaluated based on the torsional angle difference of the midpoint of the hysteresis curve width at $\pm 3\%$ of the rated torque. When the length of the swivel arm bearing roller increases, both the inclination of the hysteresis curve and the distance between its upper and lower lines change significantly. Therefore, from the change trend of the hysteresis curve shown in Fig. 21, it can be found that the length of the rollers of the swivel arm bearing has the greatest impact on the lost motion of the RV reducer. Fig. 23 illustrates the change curve of the reducer's lost motion as the roller lengths of both the support bearing and swivel arm bearing, along with the tooth surface contact widths of the involute gear and cycloid gear, increase. Through comparison, it can be seen that the swivel arm bearing has the greatest impact on the lost motion of the reducer. Increasing the length of the swivel arm bearing roller reduces the lost motion most rapidly. The length of the support bearing roller and the tooth surface contact width of the cycloid gear have relatively little effect on lost motion, while the tooth surface contact width of the involute gear has virtually no impact on lost motion.

5.2. Effect of bearing preload and clearance on torsional rigidity and lost motion

The above research indicates that the support bearing and the swivel arm bearing have a greater impact on the torsional rigidity and lost motion of the RV reducer. Preload and clearance are crucial factors that affect the stiffness performance of the support bearing and swivel arm bearing, respectively. As shown in Fig. 24, with the increase of the axial preload displacement of the supporting bearing, the inclination of the hysteresis curve decreases. When the axial preload displacement exceeds 0.03 mm, the inclination of the hysteresis curve remains almost unchanged. Furthermore, as the axial preload displacement of the support bearing increases, the distance between the upper and lower curves of the hysteresis curve decreases.

The effect of bearing preload on torsional stiffness and lost motion is shown in Fig. 25. As shown in Fig. 25(a), the torsional rigidity of the reducer increases with the increase in the axial preload displacement of the support bearing. Within the range of 0.01 mm to 0.03 mm of axial preload displacement, the torsional rigidity increases rapidly. When the axial preload displacement is <0.01 mm or greater than 0.03 mm, the torsional rigidity increases slowly. As shown in Fig. 25(b), increasing the axial preload displacement of the support bearing can significantly reduce the lost motion of the reducer. However, when the axial preload displacement exceeds 0.01 mm, further increases in the axial preload displacement have a negligible effect on reducing the lost motion.

Increasing the axial preload displacement can eliminate the clearance between the rollers and raceways and increase the bearing's load-bearing area. This can improve the bearing's load capacity, resulting in increased torsional rigidity and reduced lost motion. When the preload displacement of the supporting bearing is small, increasing the preload displacement primarily serves to eliminate the clearance between the bearing rollers and raceways, and the rollers and raceways have not formed effective contact. Therefore, when the preload displacement is <0.01 mm, the torsional rigidity increases slowly while the lost motion decreases rapidly. When the preload displacement is large enough, the contact between the rollers and raceways has essentially covered the entire load-bearing area. Additional preload displacement can no longer significantly expand the load-bearing zone. Therefore, when the preload displacement exceeds 0.03 mm, the rate of increase in the reducer's torsional rigidity slows down.

Fig. 26 shows the changes in the hysteresis curve of the RV reducer when the diameter error of the swivel arm bearing roller increases. Increased roller diameter error will increase the radial clearance of the swivel arm bearing. It can be seen that as the radial clearance increases, the inclination of the hysteresis curve gradually increases, and the distance between the upper and lower curves of the hysteresis curve increases. Besides, the area enclosed by the hysteresis curve increases. Based on hysteresis curve analysis, the statistical changes in the torsional rigidity and lost motion of the RV reducer are shown in Fig. 27. It can be found that as the radial

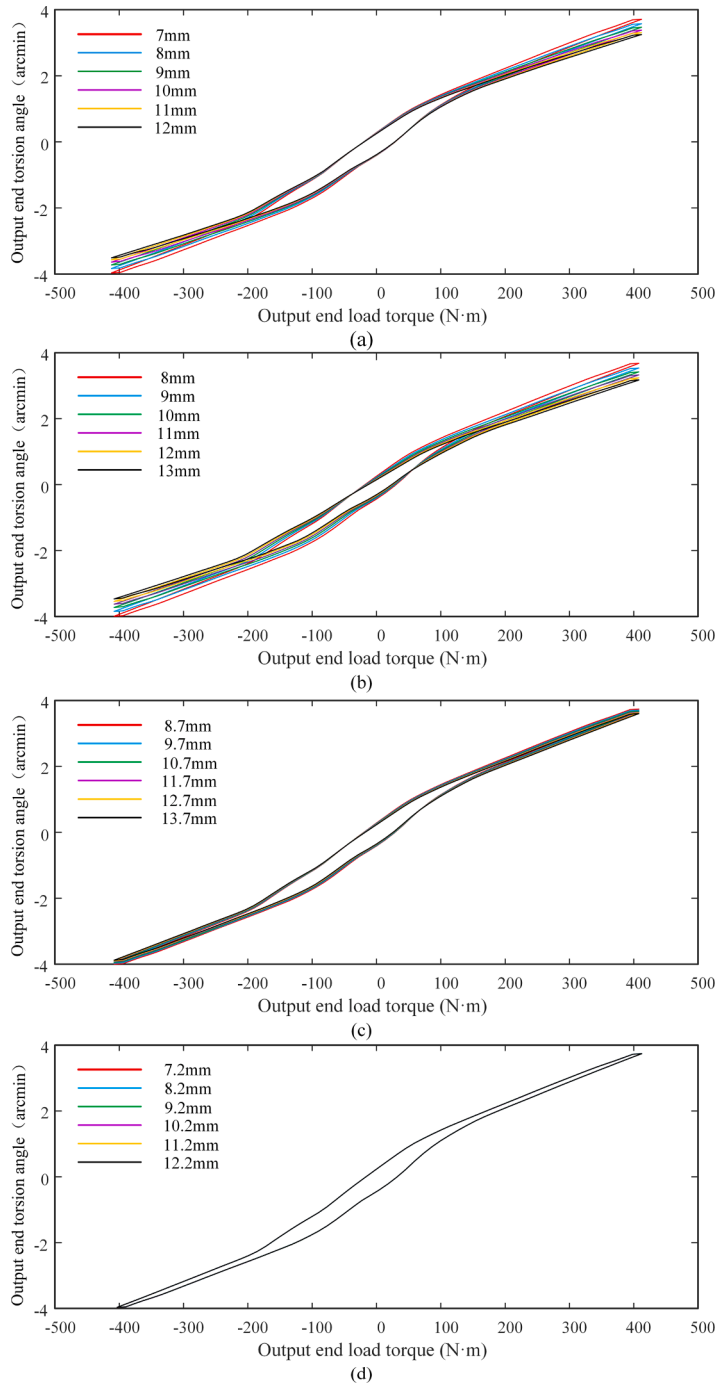


Fig. 21. Hysteresis curve change trends: (a) support bearing roller length, (b) swivel arm bearing roller length, (c) cycloid gear surface contact width, (d) involute gear tooth surface contact width.

clearance of the swivel arm bearing increases, the torsional rigidity of the RV reducer decreases by 1.2 %, while the lost motion increases by 37.81 %. It can be seen that the radial clearance of the swivel arm bearing has a significant effect on the lost motion. Torsional rigidity is evaluated within the large torque range, where the gap between the rollers and raceways has been compensated by the contact deformation, and the rollers and raceways are in close contact. Therefore, the radial clearance has a relatively small influence on the torsional rigidity. Lost motion is evaluated within a small torque range, where the gap exists between the rollers and raceways. The greater the radial clearance, the greater the gap needs to be compensated between the rollers and raceway to establish contact, resulting in a larger lag angle. Therefore, radial clearance significantly affects lost motion.

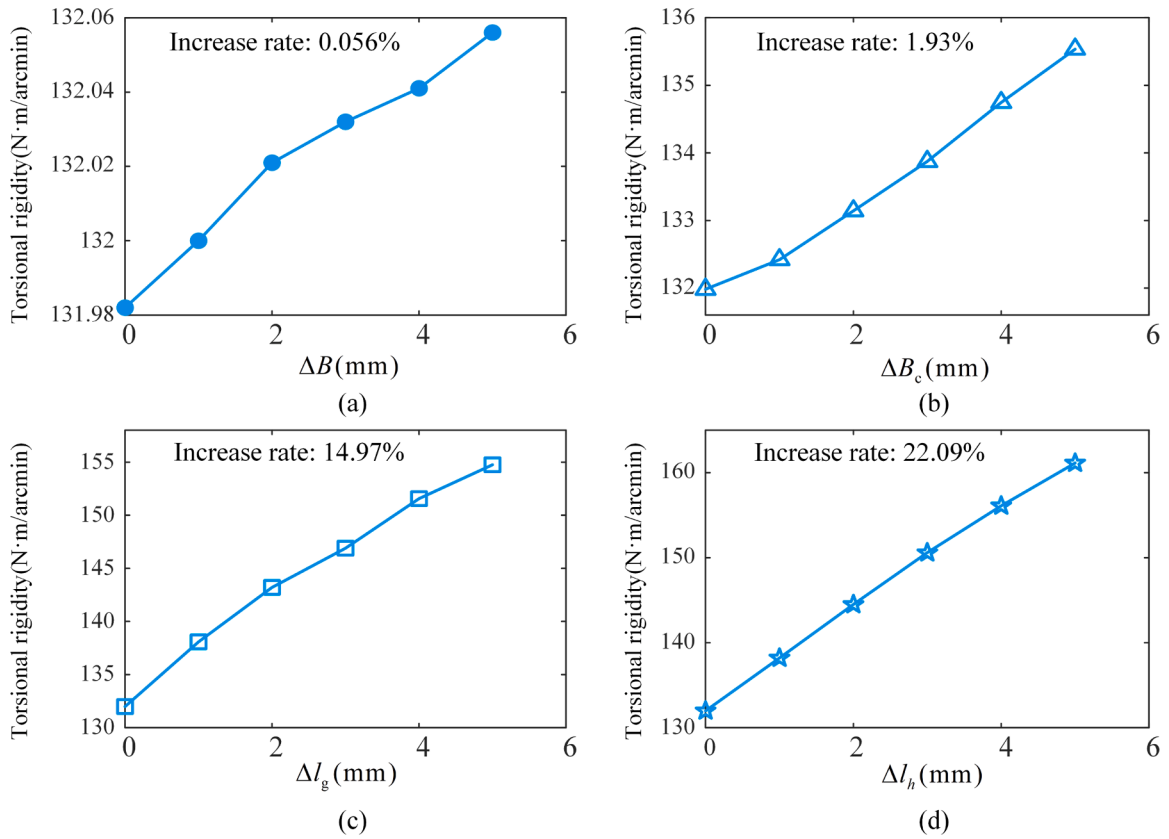


Fig. 22. Torsional rigidity change trends: (a) involute gear tooth surface contact width, (b) cycloid gear surface contact width, (c) swivel arm bearing roller length, (d) support bearing roller length.

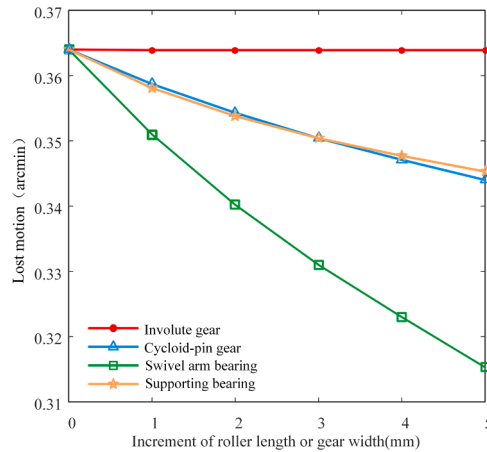


Fig. 23. Lost motion variation curve with bearing roller length and tooth surface contact width.

5.3. Effect of geometric errors of transmission components on torsional rigidity and lost motion

Geometric errors of transmission components not only affect the transmission accuracy and load capacity of the RV reducer but also have a certain impact on torsional rigidity and lost motion. Based on the manufacturing experience of the RV reducer, Table 6 presents the tolerance design of the main components of the RV reducer under three different precision grades (high, moderate, and low), and the changes in the hysteresis curve, torsional rigidity, and lost motion of the RV reducer under the three precision grades are analyzed. As shown in Fig. 28, with the decrease of geometric error accuracy, the inclination of the hysteresis curve of the RV reducer increases, while the distance between the upper and lower two curves of the hysteresis curve increases, and the area enclosed by the hysteresis

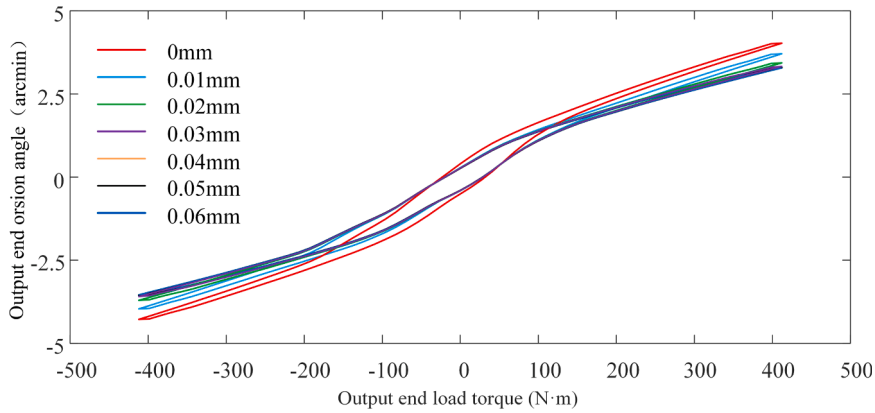


Fig. 24. Hysteresis curves under different axial preload displacements of the supporting bearing.

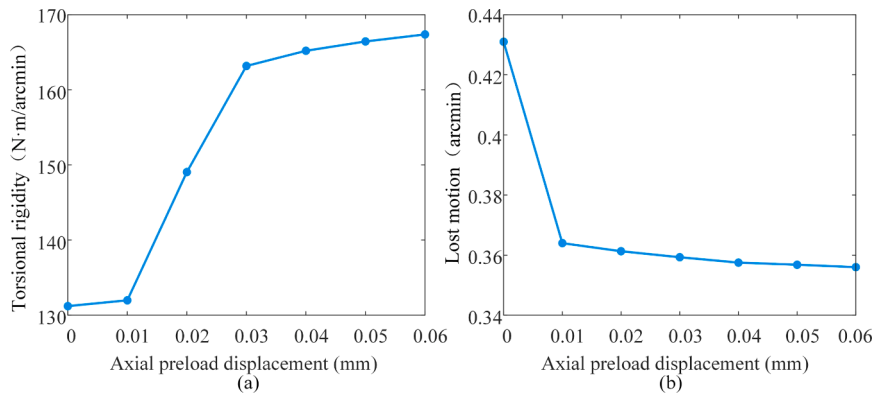


Fig. 25. Effect of bearing preload on torsional rigidity and lost motion: (a) torsional rigidity, (b) lost motion.

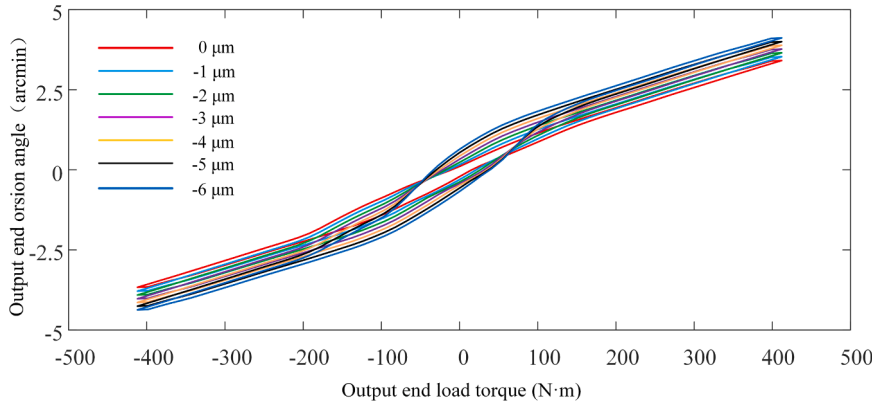


Fig. 26. Hysteresis curves under different roller diameter error of the swivel arm bearing.

curve expands. The torsional rigidity and lost motion of the reducer under the geometric errors of the three groups of precision grades are shown in Table 7. It can be found that as the geometric error accuracy decreases, the torsional rigidity of the reducer decreases, and the lost motion increases. The reduction in geometric error accuracy leads to an increase in the tooth side clearance of the cycloid-pin pair and the radial clearance of the swivel arm bearing, which results in a decrease in the torsional rigidity of the RV reducer and an increase in the lost motion.

5.4. Effect of loading rate on torsional rigidity and lost motion

The load torque loading rate can affect the hysteresis characteristics of the reducer [45]. According to the torque loading method

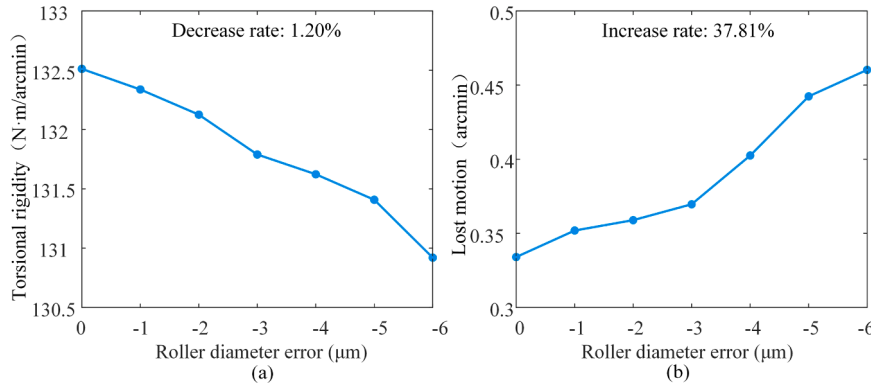


Fig. 27. Effect of roller diameter error of swivel arm bearing on torsional rigidity and lost motion: (a) torsional rigidity, (b) lost motion.

Table 6

Geometric errors design range for main transmission components of RV reducer at different precision grades.

Components	Error terms	High precision		Moderate precision		Low precision	
		Lower deviation	Upper deviation	Lower deviation	Upper deviation	Lower deviation	Upper deviation
Crankshaft	Eccentric circle radius error (μm)	-3	0	-6	0	-9	0
	Eccentric circle distribution angle error (°)	-60	+60	-90	+90	-120	+120
Cycloid gear	Eccentricity length error (μm)	-2	+2	-4	+4	-6	+6
	Cycloid gear bearing hole distribution circle radius error (μm)	-3	+3	-6	+6	-9	+9
	Cycloid gear bearing hole radius error (μm)	0	+3	0	+6	0	+9
	Cycloid gear bearing hole distribution angle error (°)	-60	+60	-90	+90	-120	+120
Pin gear	Pin tooth distribution circle radius error (μm)	0	3	0	6	0	9
	Pin tooth radius error (μm)	-3	0	-6	0	-9	0
	Pin tooth distribution angle error (°)	-30	+30	-45	+45	-60	+60
Output assembly	Supporting bearing hole distribution circle radius error (μm)	-3	+3	-6	+6	-9	+9
	Supporting bearing hole distribution angle error (°)	-60	+60	-90	+90	-120	+120

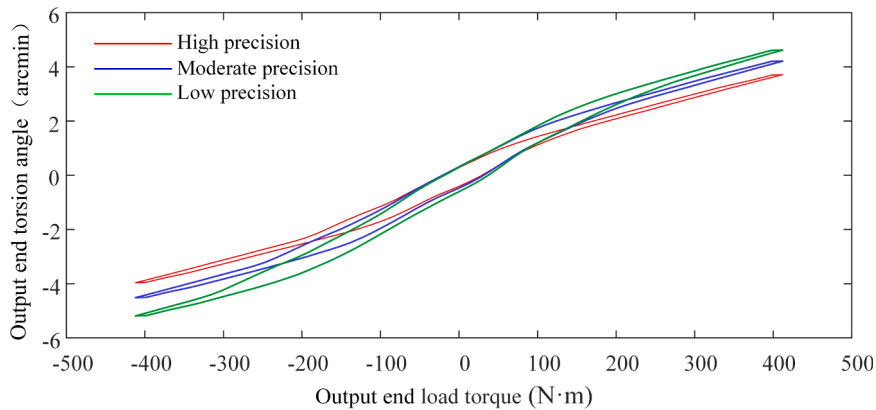


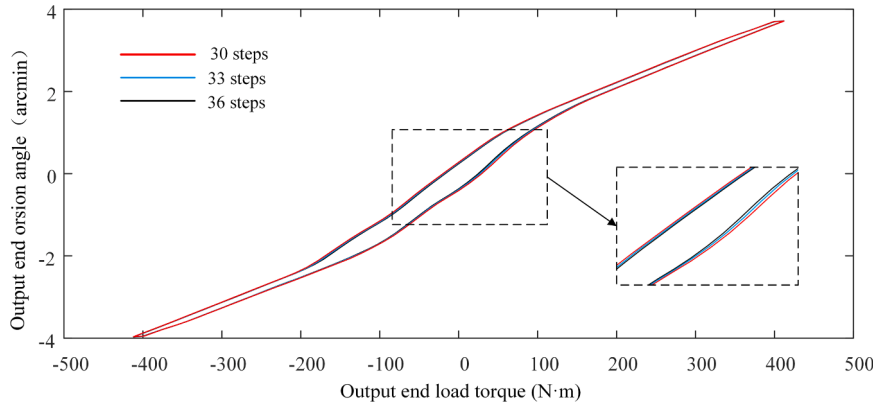
Fig. 28. Hysteresis curves for the reducer under different assembly accuracy grades.

described in Section 4.2, this paper simulates different loading rates by changing the number of loading or unloading steps for each torque segment. As shown in Fig. 29, as the number of loading or unloading steps increases, the inclination of the hysteresis curve remains almost unchanged, and the distance between the upper and lower curves of the hysteresis curve within small torque range increases. As shown in Table 8, as the number of torque loading or unloading steps increases, the torsional rigidity remains virtually

Table 7

Simulation results of torsional rigidity and lost motion of RV reducer under different assembly precision grades.

	High precision	Moderate precision	Low precision
Torsional rigidity (N·m/arcmin)	131.98	125.55	117.41
Lost motion (arcmin)	0.364	0.389	0.412

**Fig. 29.** Hysteresis curves for the reducer under different loading rates.**Table 8**

Simulation results of torsional rigidity and lost motion under different loading rates.

	30 steps	33 steps	36 steps
Torsional rigidity (N·m/arcmin)	131.98	132.11	132.22
Lost motion (arcmin)	0.364	0.337	0.314

unchanged, while lost motion is significantly reduced. It can be seen that the loading rate significantly affects lost motion; the smoother the torque loading, the smaller the lost motion. This is because the more steps are used for loading or unloading, the smoother the load torque is applied, resulting in smaller dynamic impact generated during the contact of the reducer's transmission pairs. Dynamic impact can cause additional contact deformation and increase energy loss due to friction, which in turn leads to greater lost motion. The loading rate has a negligible effect on torsional rigidity. In the high torque range, the hysteresis curve is close to linear, and all parts in the transmission system are in close contact and in a stable elastic deformation state. At this time, the torsional deformation of the reducer is mainly affected by material properties, so the loading rate has little effect on the torsional rigidity.

6. Conclusions

Based on the contact dynamic model of the RV reducer, a dynamic method is proposed for evaluating the hysteresis curve characteristics of the RV reducer. The hysteresis curve is utilized to analyze the torsional rigidity and lost motion performance of the RV reducer at different working positions. Besides, the accuracy of the analysis results is verified by the experiment. Furthermore, the effects of contact stiffness, bearing preload, radial clearance, and geometric errors on the torsional rigidity and lost motion of the reducer are explored. The main conclusions of this study are summarized as follows:

1. The roller-raceway contact stiffness of the support bearing has the most significant effect on the torsional rigidity of the RV reducer, followed by the roller-raceway contact stiffness of the swivel arm bearing. The contact stiffness of the cycloid-pin pair and the meshing stiffness of the involute gear pair have a minor impact on the torsional rigidity of the RV reducer, with the involute gear meshing stiffness having the least effect.
2. As the axial preload displacement of the support bearing increases, the inclination of the hysteresis curve decreases. When the axial preload displacement exceeds 0.03 mm, the inclination of the hysteresis curve is almost unchanged. The preload of the support bearing has a significant impact on the torsional rigidity of the RV reducer. The torsional stiffness increases significantly when the axial preload displacement ranges from 0.01 mm to 0.03 mm. As the radial clearance increases, the inclination of the hysteresis curve gradually increases, and the distance between the upper and lower curves of the hysteresis curve increases significantly. Besides, the area enclosed by the hysteresis curve increases. The radial clearance of the swivel arm bearing has a significant impact on lost motion while exerting a small effect on torsional rigidity.
3. The decrease in the geometric error accuracy grade of transmission components results in an increased inclination of the RV reducer hysteresis curve, a greater distance between the upper and lower curves of the hysteresis curve, and an enlarged area

enclosed by the hysteresis curve. Geometric errors of transmission components can significantly reduce the torsional rigidity and increase the lost motion of the RV reducer.

4. The loading rate has little effect on the inclination of the hysteresis curve of the reducer, but has a significant impact on the distance between the upper and lower curves within the small torque range. The loading rate mainly affects the lost motion of the reducer; the smoother the loading, the smaller the lost motion of the reducer.

CRedit authorship contribution statement

Lixin Xu: Methodology. **Tianyu Zhao:** Writing – original draft. **Jianwei Geng:** Validation. **Yunqing Deng:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.mechmachtheory.2025.106335](https://doi.org/10.1016/j.mechmachtheory.2025.106335).

Data availability

Data will be made available on request.

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